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GRADUATION PROJECT

Presented in view of obtaining the diploma:

Bachelor's Degree in Business Intelligence

**Design and Development of a Web Application for
telecom offer simulation and business Intelligence
Analysis**

Host Organization



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Dedications

To my dear mother,

who embodies patience, encouragement and infinite love. Your constant presence and unconditional support have been the pillars of my success.

Thank you for everything you are.

To my dear father,

This work is dedicated to you as a sign of respect and gratitude.

To my brothers and my friends,

thank you for the support and encouragement throughout these years of study. May God bring you happiness, help you achieve all your dreams and offer you a bright future.

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Table of Contents

Dedications	10
Acknowledgements.....	10
List of tables	10
List of Figures.....	10
List of Acronyms	12
ABSTRACT	10
General Introduction.....	2
CHAPTER 1: General Core of the Project	2
Introduction	3
1.1 Project scope.....	3
1.2 Presentation of the Host Organization.....	3
1.2.1 General Presentation of SOTETEL.....	3
1.3 Study of Existing Solutions.....	4
1.3.1 Description of Existing Solutions	4
1.3.2 Critique of Existing Solutions	4
1.3.3 Proposed Solution	4
1.4 Theoretical Framework and State of the Art.....	5
1.4.1 Telecom Offers (Prepaid/Postpaid, Bundles, Roaming, Fair-Use)	5
1.4.2 Tariff Simulation and Offer Recommendation Systems.....	5
1.4.3 Business Intelligence in the Telecom Sector	5
1.5 Project Management Methodology	6
1.5.1 Classification of Project Management Methodologies.....	6
1.5.2 Choice of SCRUM.....	6
1.5.3 SCRUM Fundamentals	6
1.5.4 SCRUM Adaptation for a Solo Project	7
1.5.5 Sprint Planning Overview	8
1.6 Modelling Language.....	8
Conclusion.....	8
CHAPTER 2: Sprint 0: Requirements Analysis and Specification.....	9
Summary	9

Introduction	10
2.1 Specification of Requirements.....	10
2.1.1 Actor Identification	10
2.1.2 Functional Requirements	10
2.1.3 Non-Functional Requirements.....	11
2.2 General Use Case Diagram	12
2.3 Class Diagram.....	13
2.4 Product Backlog	14
2.5 Sprint Planning.....	15
2.6 Software System Architecture.....	15
2.6.1 Backend Design (MVC Pattern)	18
2.7 Development Environment	19
2.7.1 Hardware Environment	19
2.7.2 Software Environment.....	19
Conclusion.....	26
CHAPTER 3: Sprint 1 – Core System Setup and Management Modules	27
Summary	27
Introduction	28
3.1 Sprint 1 Objectives	28
3.2 Sprint 1 Backlog	28
3.3 System Design	28
3.3.1 Use Case Diagram	29
3.3.2 Sequence Diagrams.....	29
3.3.3 Class Diagram Refinement	Error! Bookmark not defined.
3.3.4 Detailed Use Case Descriptions	34
3.4 Implementation.....	38
3.4.1 Backend Implementation.....	38
3.4.2 Frontend Implementation.....	38
3.5 Realisation	39
3.6 Testing and Validation	43
Conclusion.....	44
CHAPTER 4: Sprint 2 – Simulation Engine and Offer Comparison.....	45

Introduction	46
4.1 Sprint 2 Objectives	46
4.2 Sprint 2 Backlog	46
4.3 System Design	47
4.3.1 Use Case Diagram	47
4.3.2 Sequence Diagrams.....	48
4.3.3 Activity Diagram	49
4.3.4 Detailed Use Case Descriptions	50
4.4 Implementation.....	53
4.4.1 Simulation Engine Logic	53
4.4.2 Backend Implementation.....	53
4.4.3 Frontend Implementation.....	53
4.5 Realisation	54
4.6 Testing and Validation	59
Conclusion.....	59
CHAPTER 5: Sprint 3 – Scenario Management and Data Export	60
Introduction	61
5.1 Sprint 3 Objectives	61
5.2 Sprint 3 Backlog	61
5.3 System Design	62
5.3.1 Use Case Diagram	62
5.3.2 Sequence Diagrams.....	63
5.3.3 Detailed Use Case Descriptions	64
5.4 Implementation.....	66
5.4.1 Scenario Management Module.....	66
5.4.2 Export Functionality.....	67
5.4.3 Backend Implementation.....	67
5.4.4 Frontend Implementation.....	67
5.5 Realisation	67
5.6 Testing and Validation	72
Conclusion.....	73
CHAPTER 6: Sprint 4 – Business Intelligence and Data Visualization	74

Introduction	75
6.1 Sprint 4 Objectives	75
6.2 Sprint 4 Backlog	75
6.3 System Design	76
6.3.1 Use Case Diagram	76
6.3.2 Data Flow and Architecture.....	76
6.3.3 Detailed Use Case Descriptions	78
6.4 Implementation.....	79
6.4.1 Data Preparation	79
6.4.2 Power BI Integration	79
6.4.3 Dashboard Design	80
6.5 Realisation	81
6.5.1 Application Dashboard (Internal Analytics)	81
6.5.2 Power BI Dashboard – Marketing Analysis.....	82
6.5.3 Power BI Dashboard – Product Analysis	83
6.5.4 Power BI Dashboard – Direction and Recommendation	83
6.5.5 Interactive Features	84
6.6 Testing and Validation	85
Conclusion.....	85
GENERAL CONCLUSION AND PERSPECTIVES.....	86

List of tables

Table 2.1 product backlog.....	14
Table 2.2 Sprint planning	15
Table 3.1 sprint 1 backlog	28
Table 3.2 use case 1 login	34
Table 3.3 use case 2 logout	34
Table 3.4 use case 3 manage users	35
Table 3.5 use case 4 manage offers	35
Table 3.6 use case 5 manage customer profiles	36
Table 3.7 use case 6 password Reset.....	37
Table 3.8 testing validation	43
Table 4.1 sprint 2 backlog	46
Table 4.2 use case mono-offer simulation.....	51
Table 4.3 use case 2 multi-offer comparison.....	51
Table 4.4 use case 3 Recommendation system	52
Table 4.5 testing and validation	59
Table 5.1 Sprint 3 backlog.....	61
Table 5.2 use case 1 save scenario.....	65
Table 5.3 use case 2 Export Results	65
Table 5.4 use case 3 Duplicate scenario	66
Table 5.5 testing and validation	72
Table 6.1 Sprint 4 backlog.....	75
Table 6.2 use case 1 view dashboard.....	78
Table 6.3 use case 2 analyze Results.....	78
Table 6.4 testing and validation	85

List of Figures

Figure 1.1 company logo	3
Figure 1.2 SCRUM Process.....	7
Figure 2.1 General Use Case Diagram	12
Figure 2.2 Global Class Diagram	13
Figure 2.3 Draw.io Logo	19
Figure 2.4 Mermaid.ai Logo	20
Figure 2.5 PlantUML Logo.....	20
Figure 2.6 VS Code Logo.....	21
Figure 2.7 GitHub Logo.....	21
Figure 2.8 Node.js Logo	22
Figure 2.9 Express.js Logo	22
Figure 2.10 React Logo.....	23
Figure 2.11 Tailwind CSS Logo	23
Figure 2.12 MySQL Logo.....	24
Figure 2.13 Postman Logo	24
Figure 2.14 Swagger Logo.....	25
Figure 2.15 Docker Logo.....	25
Figure 2.16 Power BI Logo	26
Figure 2.17 Global Architecture Diagram here	26
Figure 3.1 Sprint 1 Use Case Diagram.....	Error! Bookmark not defined.
Figure 3.2 Sequence Diagram User Authentication.....	30
Figure 3.3 Sequence Diagram Create Telecom Offer.....	31
Figure 3.4 Sequence Diagram Create Customer Profile	32
Figure 3.5 Updated Class Diagram	33
Figure 3.6 Login Interface.....	40
Figure 3.7 User Management	41
Figure 3.8 Offers Management.....	42
Figure 3.9 Customer Profiles	43
Figure 4.1 Sprint 2 Use Case Diagram.....	47
Figure 4.2 Mono-Offer Simulation Sequence Diagram	48
Figure 4.3 Multi-Offer Comparison Sequence Diagram	49
Figure 4.4 Simulation Activity Diagram	50
Figure 4.5 Simulation Main Page.....	54
Figure 4.6 Single Simulation	55
Figure 4.7 Comparison Simulation.....	56
Figure 4.8 Recommendation	57
Figure 4.9 Batch Simulation	58
Figure 5.1 Sprint 3 Use Case Diagram.....	62
Figure 5.2 Save Scenario Sequence Diagram.....	63

Figure 5.3 Export Results Sequence Diagram 64

Figure 5.4 Save Scenario 68

Figure 5.5 Scenario List 69

Figure 5.6 Scenario Details 70

Figure 5.7 Export Results 71

Figure 5.8 Export Results 71

Figure 6.1 Sprint 4 Use Case Diagram..... 76

Figure 6.2 Data Flow Diagram 77

Figure 6.3 Application Dashboard 81

Figure 6.4 Power BI Marketing Dashboard 82

Figure 6.5 Power BI Product Dashboard..... 83

Figure 6.6 Power BI Direction Dashboard 83

Figure 6.7 Power BI Filtering 84

List of Acronyms

Acronym	Meaning
API	Application Programming Interface
ARPU	Average Revenue Per User
BI	Business Intelligence
CRUD	Create, Read, Update, Delete
CSS	Cascading Style Sheets
DAX	Data Analysis Expressions
ER	Entity-Relationship
HTML	HyperText Markup Language
JWT	JSON Web Token
KPI	Key Performance Indicator
MVC	Model-View-Controller
RBAC	Role-Based Access Control
REST	Representational State Transfer
RGPD	Règlement Général sur la Protection des Données
SQL	Structured Query Language
UML	Unified Modeling Language
UI	User Interface
UX	User Experience

ABSTRACT

This graduation project presents the design and development of a Telecom Offers Simulation and Business Intelligence Analysis Platform, carried out during an end-of-studies internship at SOTETEL. The platform allows marketing and product teams to simulate the total cost and customer satisfaction of various telecom offers (voice, data, SMS, convergent packs) based on realistic customer consumption profiles. It includes an intelligent multi-offer comparator with ranking and explainability, full scenario management, and automatic export of simulation data to Power BI for the generation of decision-making dashboards.

Key features include catalogue management, customer profile modelling, real-time cost and satisfaction calculations, and advanced BI indicators such as simulated ARPU, overage rates, churn risk, and competitiveness heatmaps. The solution was built using modern technologies: React (Vite) for the front-end, Node.js/Express with mysql2 for the back-end, MySQL database, and Power BI Desktop for analytics. A dataset of 300 fictitious customer profiles was used for testing and validation.

The application aims to optimise offer selection, accelerate marketing analysis, and provide actionable insights for SOTETEL's commercial strategy.

Keywords: *Telecom Offer Simulation, Business Intelligence, Power BI, React, Node.js, Tariff Modelling, Customer Profiling, ARPU, Churn Analysis.*

General Introduction

In today’s highly competitive telecommunications landscape, operators must constantly refine their commercial offers to attract and retain customers. The growing diversity of telecom packages — spanning prepaid, postpaid, data-only, and convergent bundles — combined with complex pricing structures involving quotas, fair-use policies, and overage charges, makes manual offer comparison extremely challenging for marketing and product teams.

This internship at SOTETEL, a leading Tunisian telecommunications infrastructure company, gave us the opportunity to address this challenge by designing and developing a web application called “Telecom Offers Simulation and Business Intelligence Analysis Platform.” The platform enables users to simulate telecom offers against realistic customer consumption profiles, compare multiple offers with intelligent ranking, manage simulation scenarios, and generate actionable Business Intelligence dashboards through Power BI integration.

This report is structured into the following chapters:

- Chapter 1 presents the general framework of the project, including the host organisation, the theoretical background, the methodology adopted, and the modelling language.
- Chapter 2 (Sprint 0) is dedicated to requirements analysis and specification, covering functional and non-functional requirements, the general use case diagram, the class diagram, the product backlog, sprint planning, the development environment, and the system architecture.
- Chapter 3 (Sprint 1) details the implementation of core system setup and management modules, including authentication, user management, offers management, and customer profiles management.
- Chapter 4 (Sprint 2) explains the development of the simulation engine and offer comparison functionalities.
- Chapter 5 (Sprint 3) covers scenario management and data export capabilities.
- Chapter 6 (Sprint 4) presents the Business Intelligence module and data visualisation through Power BI dashboards.

Finally, a general conclusion summarises the achievements and presents perspectives for future work.

CHAPTER 1: General Core of the Project

Summary

Introduction	3
1.1 Project scope.....	3
1.2 Presentation of the Host Organization.....	3
1.2.1 General Presentation of SOTETEL.....	3
1.3 Study of Existing Solutions.....	4
1.3.1 Description of Existing Solutions	4
1.3.2 Critique of Existing Solutions	4
1.3.3 Proposed Solution	4
1.4 Theoretical Framework and State of the Art.....	5
1.4.1 Telecom Offers (Prepaid/Postpaid, Bundles, Roaming, Fair-Use)	5
1.4.2 Tariff Simulation and Offer Recommendation Systems.....	5
1.4.3 Business Intelligence in the Telecom Sector	5
1.5 Project Management Methodology	6
1.5.1 Classification of Project Management Methodologies.....	6
1.5.2 Choice of SCRUM.....	6
1.5.3 SCRUM Fundamentals	6
1.5.4 SCRUM Adaptation for a Solo Project	7
1.5.5 Sprint Planning Overview	8
1.6 Modelling Language.....	8
Conclusion.....	8

Introduction

This first chapter immerses us in the general context of the project. First, we present the project scope and the host organisation. Then, we examine the theoretical framework and state of the art in telecom offer simulation and Business Intelligence. Next, we discuss the study of existing solutions, the methodology adopted, and the modelling language. This foundational chapter sets the stage for the subsequent development sprints.

1.1 Project scope

This internship is part of the validation of our end-of-study internship to obtain the Bachelor's degree in Business Intelligence at the Faculty of Economic and Management Sciences of Tunis (FSEGT), University of Tunis El Manar. We were welcomed within SOTETEL for a four-month internship period of 16 weeks.

The platform allows the simulation of the total cost and customer satisfaction of various telecom offers (prepaid, postpaid, convergent packs) based on realistic customer consumption profiles. It includes an intelligent comparator, scenario management, and export to Power BI for strategic decision-making.

1.2 Presentation of the Host Organization

1.2.1 General Presentation of SOTETEL

SOTETEL (Société Tunisienne d'Entreprises de Télécommunications) was founded in 1981 and has established itself as a major player in the Tunisian telecommunications infrastructure sector. The company operates as a “Smart Enabler,” providing end-to-end services in the design, implementation, and maintenance of advanced telecommunication networks.

Its core activities include deployment and maintenance of radio and GSM networks, optical fibre infrastructure (more than 30,000 km deployed), installation and management of over 3,800 telecom towers, data centre solutions, cybersecurity, and Internet of Things (IoT) and Smart City applications. SOTETEL plays a unique role as an infrastructure specialist rather than a direct retail operator, enabling the rapid rollout of 4G/5G services and supporting Tunisia's national digital transformation strategy.



Figure 1.1 company logo

1.3 Study of Existing Solutions

1.3.1 Description of Existing Solutions

Before the development of this platform, telecom operators in Tunisia typically managed their offer analysis using manual processes. Marketing teams relied on spreadsheet-based calculations (Excel) to compare offers, which involved manually entering tariff parameters, consumption data, and pricing rules for each customer profile. Some operators used basic internal tools with limited simulation capabilities, but these were generally confined to single-offer evaluation without multi-criteria comparison or intelligent recommendation.

1.3.2 Critique of Existing Solutions

The analysis of existing offer management and simulation practices revealed the following deficiencies:

- **Lack of Automation:** Manual calculations are time-consuming, error-prone, and do not scale when hundreds of customer profiles need to be evaluated against dozens of offers.
- **No Multi-Offer Comparison:** Existing tools do not support simultaneous comparison of multiple offers with ranking and explainability.
- **Absence of Centralised Tracking:** There is no unified platform to save, retrieve, and duplicate simulation scenarios for reuse.
- **Limited BI Integration:** Simulation results are not automatically connected to Business Intelligence tools, preventing the generation of real-time dashboards and KPI tracking.
- **No Satisfaction Scoring:** Existing approaches focus solely on cost without considering customer satisfaction metrics such as fair-use penalties, budget alignment, or segment-specific bonuses.

1.3.3 Proposed Solution

To address the challenges outlined above, we propose the development of a comprehensive web-based Telecom Offers Simulation and BI Analysis Platform with the following objectives:

- **Automate Tariff Simulation:** Enable real-time calculation of total cost and satisfaction scores for any combination of customer profile and telecom offer.
- **Multi-Offer Comparison:** Allow side-by-side comparison of multiple offers with ranking by cost and satisfaction, including detailed justification.
- **Intelligent Recommendation:** Automatically suggest the most suitable offers based on customer profiles using multi-criteria decision-making.

- **Scenario Management:** Allow users to save, duplicate, and export simulation scenarios for later analysis.
- **BI Integration:** Provide seamless export to Power BI with pre-built dashboards for marketing, product, and executive decision-making.

1.4 Theoretical Framework and State of the Art

1.4.1 Telecom Offers (Prepaid/Postpaid, Bundles, Roaming, Fair-Use)

The Tunisian telecom market offers a wide variety of commercial packages designed to meet different customer needs. These offers are typically divided into two main categories: prepaid (pay-as-you-go) and postpaid (monthly billing). Both categories include voice minutes, SMS, and mobile data, with increasing popularity of convergent packs that combine fixed and mobile services.

Key components of modern telecom offers include quotas (fixed allowances for voice, SMS, and data), validity periods, fair-use policies (limits on excessive consumption), out-of-bundle rates (premium pricing beyond quotas), options and add-ons (roaming, night data, social bundles, loyalty discounts), and promotions tied to engagement periods. These complex rules make manual comparison extremely difficult for marketing teams [itu2023].

1.4.2 Tariff Simulation and Offer Recommendation Systems

Tariff simulation is a well-established practice in the telecom industry [bertin2020telecom]. It consists of calculating the total cost an offer would generate for a specific customer profile by applying the following logic:

- Included cost = monthly subscription price
- Overage minutes = $\max(0, \text{average minutes} - \text{quota minutes}) \times \text{out-of-bundle rate}$
- Overage SMS = $\max(0, \text{average SMS} - \text{quota SMS}) \times \text{out-of-bundle rate}$
- Overage data = $\max(0, \text{average data (GB)} - \text{quota data}) \times \text{out-of-bundle rate}$
- Total cost = Included cost + Overages – Discounts/Promotions

In addition to cost, modern systems calculate a satisfaction score based on coverage of needs, penalties for fair-use violations, and alignment with customer priorities (price vs. quality). Recommendation engines then rank multiple offers using multi-criteria decision-making methods, providing explainability.

1.4.3 Business Intelligence in the Telecom Sector

Business Intelligence (BI) plays a strategic role in telecom operators by transforming raw simulation and consumption data into actionable insights [kimball2013datawarehouse]. The most important KPIs include ARPU (Average Revenue Per User), churn risk (probability of customer departure due to high overage costs), overage rate (percentage of customers exceeding quotas),

offer competitiveness (comparison against market benchmarks), and segment performance (analysis by prepaid/postpaid, budget level, or usage type).

Power BI is widely used in the industry because of its powerful DAX language, interactive dashboards, and ability to connect directly to exported datasets [microsoft2024powerbi]. Typical dashboards include heatmaps of overages, trend analysis of simulated ARPU, and scenario comparison views for marketing and executive decision-making.

1.5 Project Management Methodology

1.5.1 Classification of Project Management Methodologies

Project management methodologies enable the planning and implementation of projects over a specific period. These methodologies are divided into two types [denjean2004conduite]:

- **Classical Methods:** Also known as traditional or waterfall methods, they are characterised by a succession of phases, each linked to the previous one. To move to the next stage, the preceding one must be completed.
- **Agile Methods:** More flexible and less rigid than traditional methods, agile approaches focus on the customer's needs. They follow an iterative and collaborative approach which takes into account both the initial requirements and those linked to the evolution of the system.

1.5.2 Choice of SCRUM

In our project, we adopted the SCRUM agile methodology, as it is the most appropriate for this type of iterative development. SCRUM offers several advantages [sutherland2014scrum]:

- **Risk reduction:** The project is broken down into short iterations, making problems easier to detect.
- **Visibility:** We can see how the project is progressing, assess obstacles, and track tasks.
- **Correct planning:** Long-term global planning combined with detailed short-term planning.

1.5.3 SCRUM Fundamentals

The SCRUM methodology is based on three pillars: transparency (all aspects of the process must be visible), inspection (frequent checks of progress towards the sprint goal), and adaptation (adjustments when deviations are detected) [schwaber2020scrumguide].

SCRUM defines three key roles:

- **Product Owner:** Provides the link between the customer and the development team by defining clear and achievable requirements. Responsible for approving or rejecting work.
- **Scrum Master:** Ensures that the SCRUM method is implemented correctly and helps the team adapt to its practices.

- Development Team: Responsible for transforming the requirements in the Product Backlog into concrete functionalities. The team is self-organised and multi-disciplinary.

Key SCRUM terminology includes:

- Product Backlog: A prioritised list of product functionalities representing all tasks to be accomplished.
- Sprint Backlog: The list of functionalities to be implemented during each sprint, established during sprint planning.
- Sprint Planning: A meeting to establish the next sprint's objectives and scope.
- Daily Scrum: A 15-minute daily meeting to share progress, specify tasks, and identify obstacles.
- Sprint Review: A meeting to publicly showcase outcomes and collect stakeholder feedback.

1.5.4 SCRUM Adaptation for a Solo Project

Since this project was developed by a single developer, the SCRUM roles were adapted as follows:

- Product Owner: Mr. Mourad Ben Mna (with the student)
- Scrum Master: The student
- Development Team: Ghaith bejaoui

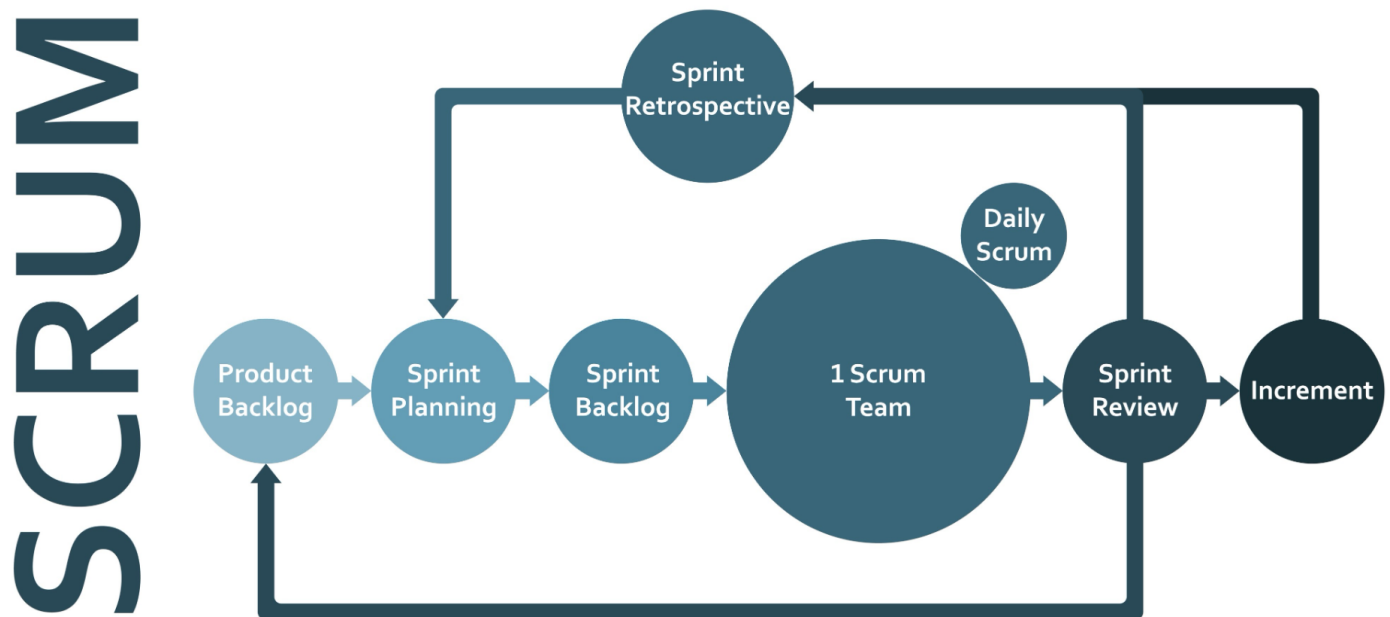


Figure 1.2 SCRUM Process

1.5.5 Sprint Planning Overview

Sprint	Duration	Main Objectives	Status
Sprint 0	Weeks 1–2	Requirements, architecture, UML diagrams	Completed
Sprint 1	Weeks 3–6	Core system: authentication, CRUD modules	Completed
Sprint 2	Weeks 7–10	Simulation engine, comparison, recommendation	Completed
Sprint 3	Weeks 11–13	Scenario management, export functionalities	Completed
Sprint 4	Weeks 14–16	BI integration, Power BI dashboards, testing	Completed

1.6 Modelling Language

The Unified Modeling Language (UML) was created to forge a common, semantically and syntactically rich visual modelling language for the architecture, design, and implementation of complex software systems both structurally and behaviourally [omg2017uml]. UML was used throughout this project for use case diagrams, sequence diagrams, class diagrams, and activity diagrams.

Conclusion

In this chapter, we situated ourselves within the general framework of the project. We presented the host organisation, examined the state of the art and existing solutions, detailed the proposed solution, and explained the project management methodology. This solid foundation established in Sprint 0 (Chapter 2) allowed the smooth execution of the subsequent development sprints.

CHAPTER 2:

Sprint 0: Requirements Analysis and Specification

Summary

- Introduction 10
- 2.1 Specification of Requirements..... 10
 - 2.1.1 Actor Identification 10
 - 2.1.2 Functional Requirements 10
 - 2.1.3 Non-Functional Requirements..... 11
- 2.2 General Use Case Diagram 12
- 2.3 Class Diagram..... 13
- 2.4 Product Backlog 14
- 2.5 Sprint Planning..... 15
- 2.6 Software System Architecture..... 15
 - 2.6.1 Backend Design (MVC Pattern) 18
- 2.7 Development Environment 19
 - 2.7.1 Hardware Environment 19
 - 2.7.2 Software Environment..... 19
- Conclusion..... 26

Introduction

Sprint 0 represents the foundational phase of the project, dedicated to requirements analysis and specification. Its primary objective is to clearly identify and formalise both functional and non-functional requirements of the Telecom Offers Simulation and Business Intelligence Platform developed for SOTETEL. This phase also includes system modelling using UML diagrams, the establishment of the product backlog, and the definition of the technical environment and global system architecture.

2.1 Specification of Requirements

2.1.1 Actor Identification

The system involves three main categories of users:

Actor	Description
Administrator	Has full access to the system. Responsible for managing users, telecom offers, options, and accessing audit logs and system configurations.
Analyst	Uses the platform to manage telecom offers and customer profiles, execute simulations, create scenarios, and analyse Business Intelligence dashboards.
Guest	Has restricted access. Can view available offers and customer profiles and run simulations, but cannot save scenarios or export results.

2.1.2 Functional Requirements

The functional requirements are structured into the following main modules:

- Catalogue Management: CRUD operations on telecom offers with lifecycle management (Published, Draft, Retired).
- Options Management: Management of add-ons (roaming, data bundles, promotional options) and their association with offers.
- Customer Profiles Management: Creation and management of customer usage profiles based on consumption patterns, budget constraints, and preferences.
- Simulation Engine: Single-offer simulation, multi-offer comparison, intelligent recommendation, and batch simulations.
- Scenario Management: Save, duplicate, manage, and export simulation scenarios.

- Business Intelligence Module: Automatic export of simulation data to Power BI and generation of analytical dashboards.
- Audit and Traceability: Full tracking of user actions for monitoring and compliance.

2.1.3 Non-Functional Requirements

The system must satisfy the following quality attributes:

- Security:
Implementation of Role-Based Access Control (RBAC), JWT authentication, password encryption (bcrypt), and protection against common vulnerabilities.
- Performance:
The system must ensure fast response times, with simulations executed in less than 2 seconds, even in batch mode.
- Usability:
The application should provide an intuitive, user-friendly, and responsive interface adaptable to multiple devices.
- Scalability:
The system must support a growing number of offers, users, and simulation operations
- Traceability:
All system actions must be logged and traceable through an audit mechanism.
- Data Governance:
Compliance with data protection standards (RGPD), including anonymization of customer profiles.
- Maintainability:
The codebase should be modular, well-documented, and easy to maintain or extend.

2.2 General Use Case Diagram

The general use case diagram illustrates the interactions between system actors and the platform's main functionalities.

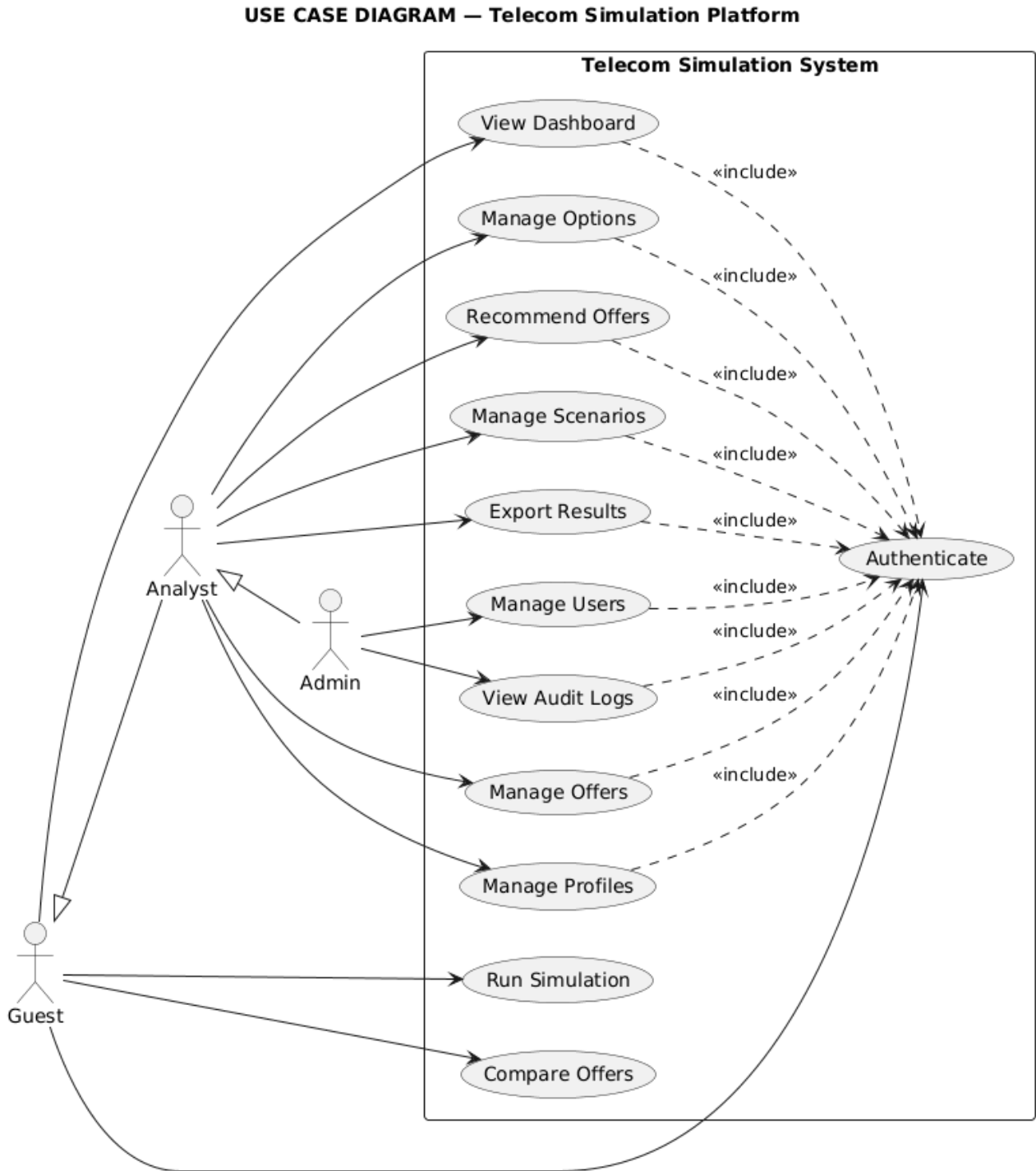


Figure 2.1 General Use Case Diagram

2.3 Class Diagram

The class diagram represents the conceptual structure of the system, defining entities, attributes, and relationships. The main entities include: User, Offer, Option, OfferOption, CustomerProfile, Scenario, ScenarioResult, FactSimulation, and AuditLog.

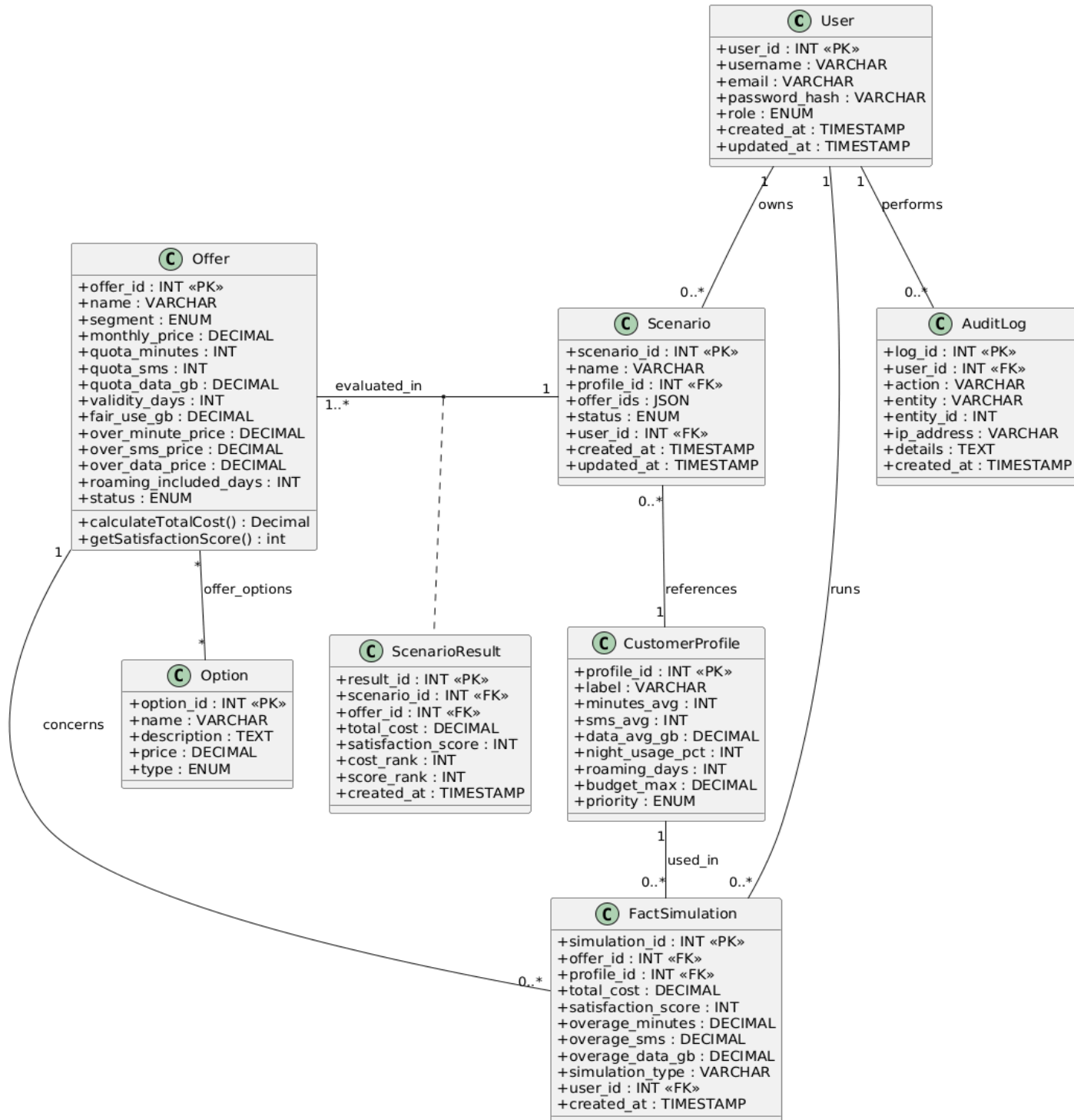


Figure 2.2 Global Class Diagram

2.4 Product Backlog

The product backlog gathers all user stories prioritised according to their business value.

Table 2.1 product backlog

ID	Functionality	User Story	Priority	Sprint
US01	Authentication	As a user, I need to authenticate to access the platform	High	1
US02	Manage Users	As an admin, I can manage user accounts	High	1
US03	Manage Offers	As an analyst, I can manage telecom offers (CRUD)	High	1
US04	Manage Profiles	As an analyst, I can manage customer profiles	High	1
US05	Manage Options	As an analyst, I can manage offer options	High	1
US06	Mono-Offer Simulation	As an analyst, I can simulate a single offer	High	2
US07	Multi-Offer Comparison	As an analyst, I can compare multiple offers	High	2
US08	Recommendation	As an analyst, I get intelligent offer recommendations	High	2
US09	Batch Simulation	As an analyst, I can run batch simulations	High	2
US10	Manage Scenarios	As an analyst, I can save and manage scenarios	Medium	3
US11	Export Results	As an analyst, I can export simulation results	Medium	3
US12	Consult BI Dashboards	As an analyst, I can view BI dashboards	High	4
US13	Audit Logs	As an admin, I can view audit logs	Medium	4

2.5 Sprint Planning

The project was planned over 16 weeks, divided into 8 sprints of 2 weeks each.

Table 2.2 Sprint planning

Sprint	Duration	Objectives	Key Deliverables
Sprint 0	Weeks 1–2	Requirements and architecture	Specifications, UML diagrams, database schema
Sprint 1	Weeks 3–6	Core system setup	Authentication, CRUD modules, database
Sprint 2	Weeks 7–10	Simulation engine	4 simulation modes, comparison, recommendation
Sprint 3	Weeks 11–13	Scenarios and export	Scenario CRUD, CSV/XLSX export
Sprint 4	Weeks 14–16	BI and polishing	Power BI dashboards, audit, testing

2.6 Software System Architecture

The platform follows a four-layer architecture:

- **Presentation Layer (React):** A modern JavaScript library used to build dynamic and responsive user interfaces. It enables efficient rendering and a smooth user experience through reusable components.
- **Application Layer (Node.js / Express):** Node.js provides a scalable server-side runtime environment, while Express is a lightweight framework used to build RESTful APIs and handle business logic efficiently.
- **Data Layer (MySQL):** A relational database management system used to store and manage structured data such as telecom offers, customer profiles, and simulation results.
- **Analytics Layer (Power BI):** A business intelligence tool used to visualize data through interactive dashboards and reports, supporting decision-making processes.

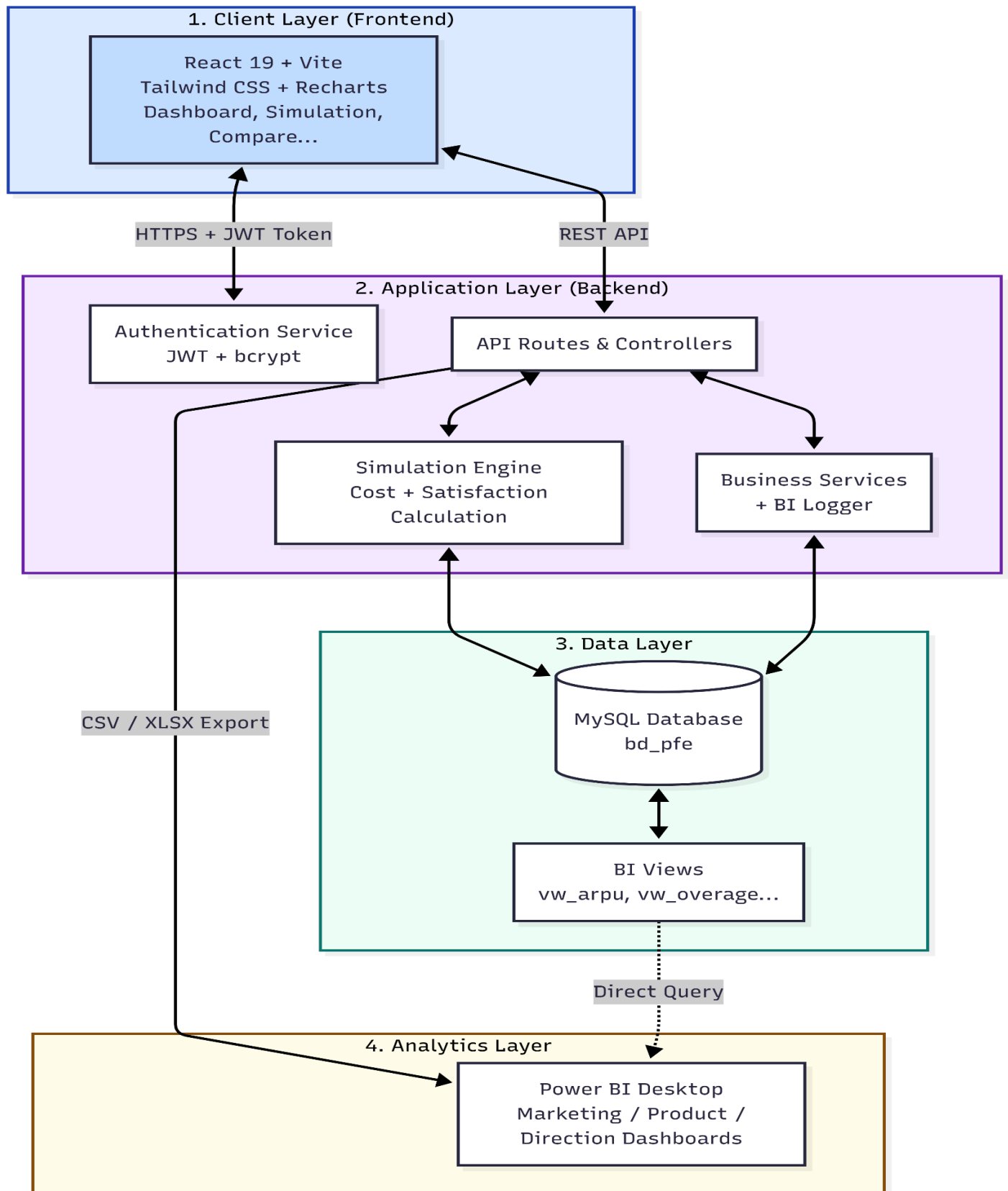


Figure 2.3 Global Architecture Diagram

Figure 2.3 illustrates the global architecture of the platform, showing the interaction between the frontend, backend, database, and Business Intelligence layers.

2.6.1 Backend Design (MVC Pattern)

The backend architecture of the platform is based on the Model-View-Controller (MVC) design pattern, which ensures a clear separation of concerns and improves maintainability, scalability, and testability of the application.

- **Models:**
The Models represent the data layer of the application. They are responsible for interacting with the database (MySQL) and managing data persistence. Models define the structure of entities such as users, telecom offers, options, and customer profiles. They also handle data validation, queries, and relationships between different entities, ensuring data integrity and consistency across the system.
- **Controllers:**
Controllers act as the core of the application's business logic. They receive incoming requests from the client, process them, and coordinate with the Models to perform the required operations. Controllers handle tasks such as creating, updating, retrieving, and deleting data, as well as executing simulation logic and managing scenarios. They also prepare and structure the responses sent back to the client.
- **Routes/Services:**
Routes define the API endpoints exposed by the backend and map HTTP requests (GET, POST, PUT, DELETE) to the appropriate controller functions. Services are used to encapsulate complex business logic and reusable functionalities, such as simulation algorithms or data processing tasks. This separation enhances code modularity and allows easier maintenance and scalability of the system.

2.7 Development Environment

2.7.1 Hardware Environment

Component	Specification
Processor	Intel Core i5-12400F
RAM	16 GB
Storage	SSD 512 GB
Operating System	Windows 11

2.7.2 Software Environment

The development of the Telecom Offers Simulation and Business Intelligence Platform relied on a set of modern tools and technologies, each selected to ensure efficiency, scalability, and maintainability.

- Design Tools:
 - Draw.io
Used for creating UML diagrams such as use case diagrams, class diagrams, and system architecture representations.



Figure 2.4 Draw.io Logo

- Mermaid.ai
Utilized for generating diagrams through code, allowing quick visualization of workflows and system structures.



Figure 2.5 Mermaid.ai Logo

- PlantUML Online
Used to design UML diagrams using a textual description approach, facilitating version control and easy updates.



Figure 2.6 PlantUML Logo

- Development Tools:

- Visual Studio Code (VS Code)
A lightweight and powerful code editor used for writing, debugging, and managing the project codebase.

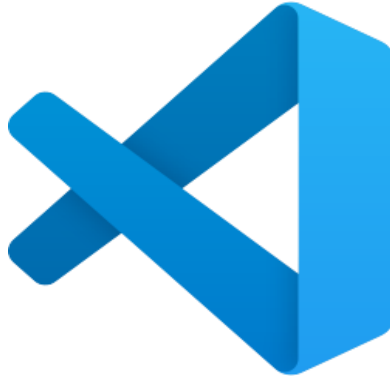


Figure 2.7 VS Code Logo

- GitHub
Used for version control, collaboration, and maintaining the project repository.

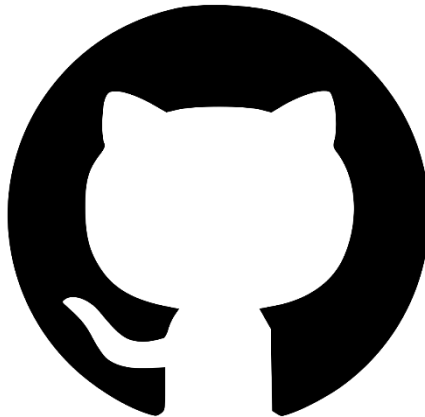


Figure 2.8 GitHub Logo

- Backend Technologies:
 - Node.js
A JavaScript runtime environment used to build scalable server-side applications.



Figure 2.9 Node.js Logo

- Express.js
A minimal and flexible Node.js web application framework used to develop RESTful APIs.



Figure 2.10 Express.js Logo

- Frontend Technologies:
 - React (Vite)
A JavaScript library used to build dynamic and responsive user interfaces, with Vite providing fast development and build performance.

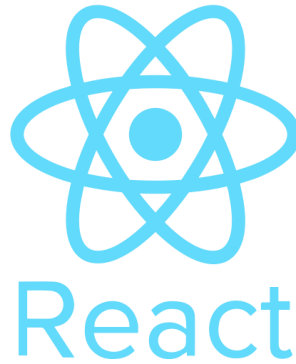


Figure 2.11 React Logo

- Tailwind CSS
A utility-first CSS framework used to design modern and responsive user interfaces efficiently.



Figure 2.12 Tailwind CSS Logo

- Database:
 - MySQL (mysql2)
A relational database management system used to store and manage application data, with mysql2 providing efficient Node.js integration.



Figure 2.13 MySQL Logo

- Testing Tools:
 - Postman
Used for testing and validating API endpoints during development.



Figure 2.14 Postman Logo

- Swagger
Used for API documentation and interactive testing, allowing clear visualization and validation of RESTful services.



Figure 2.15 Swagger Logo

- Containerization:
 - Docker
Used to containerize the application, ensuring consistency across development and deployment environments as required by the company.

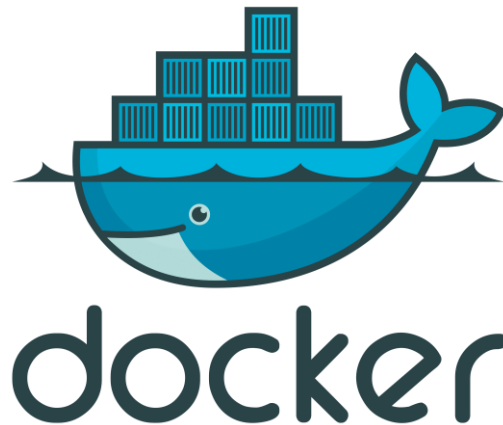


Figure 2.16 Docker Logo

- BI Tool:
 - Power BI Desktop
Used to create interactive dashboards and perform advanced data analysis for decision-making support.



Figure 2.17 Power BI Logo

Figure 2.18 Global Architecture Diagram here

Conclusion

Sprint 0 established a comprehensive foundation for the project by defining requirements, selecting the development methodology, and designing the system architecture. This preparation phase significantly reduces development risks and ensures the smooth progression of subsequent sprints.

CHAPTER 3: Sprint 1 – Core System Setup and Management Modules

Summary

Introduction	28
3.1 Sprint 1 Objectives	28
3.2 Sprint 1 Backlog	28
3.3 System Design	28
3.3.1 Use Case Diagram	29
3.3.2 Sequence Diagrams.....	29
3.3.3 Class Diagram Refinement	Error! Bookmark not defined.
3.3.4 Detailed Use Case Descriptions	34
3.4 Implementation.....	38
3.4.1 Backend Implementation.....	38
3.4.2 Frontend Implementation.....	38
3.5 Realisation	39
3.6 Testing and Validation	43
Conclusion.....	44

Introduction

Sprint 1 represents the first development phase of the project. It focuses on setting up the core system architecture and implementing the fundamental management modules required for the platform's operation.

The main goal of this sprint is to establish a functional backend and frontend structure, while developing essential features such as user management, telecom offers management, and customer profiles management. These components form the backbone of the system and are necessary for enabling more advanced functionalities in subsequent sprints.

3.1 Sprint 1 Objectives

The primary objectives of Sprint 1 are:

- Set up the development environment and project structure
- Implement authentication and authorisation mechanisms (JWT + bcrypt)
- Develop user management functionalities (CRUD + RBAC)
- Implement telecom offers management with lifecycle (Published/Draft/Retired)
- Develop options management and offer-option association
- Implement customer profiles management
- Establish database schema and relationships

3.2 Sprint 1 Backlog

Table 3.1 sprint 1 backlog

ID	User Story	Description	Priority
US01	Authentication	As a user, I want to authenticate securely to access the platform	High
US02	Manage Users	As an administrator, I want to manage users and roles	High
US03	Manage Offers	As an analyst, I want to manage telecom offers	High
US04	Manage Options	As an analyst, I want to manage telecom options	High
US05	Manage Profiles	As an analyst, I want to manage customer profiles	High

3.3 System Design

3.3.1 Use Case Diagram

This diagram illustrates the interactions between the administrator and analyst with the system during Sprint 1 functionalities.

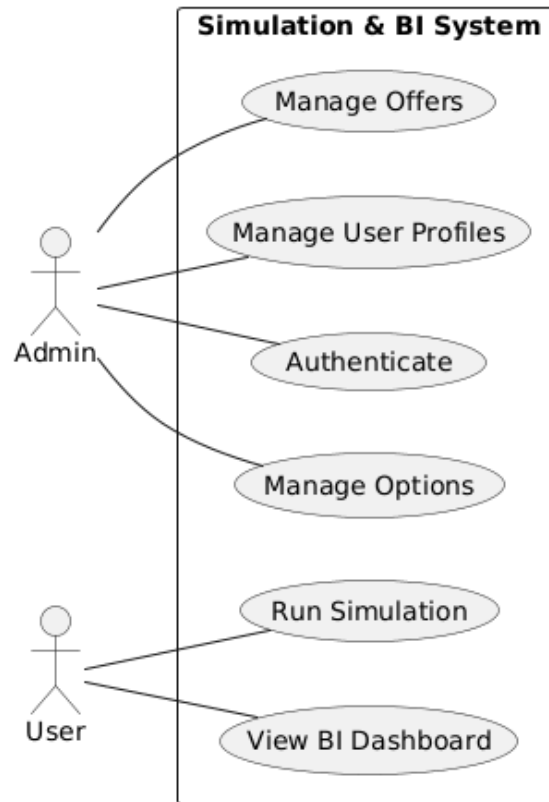


Figure 3.1 Sprint 1 Use Case Diagram

3.3.2 Sequence Diagrams

Sequence diagrams describe the interaction between system components over time.

- User Authentication Sequence

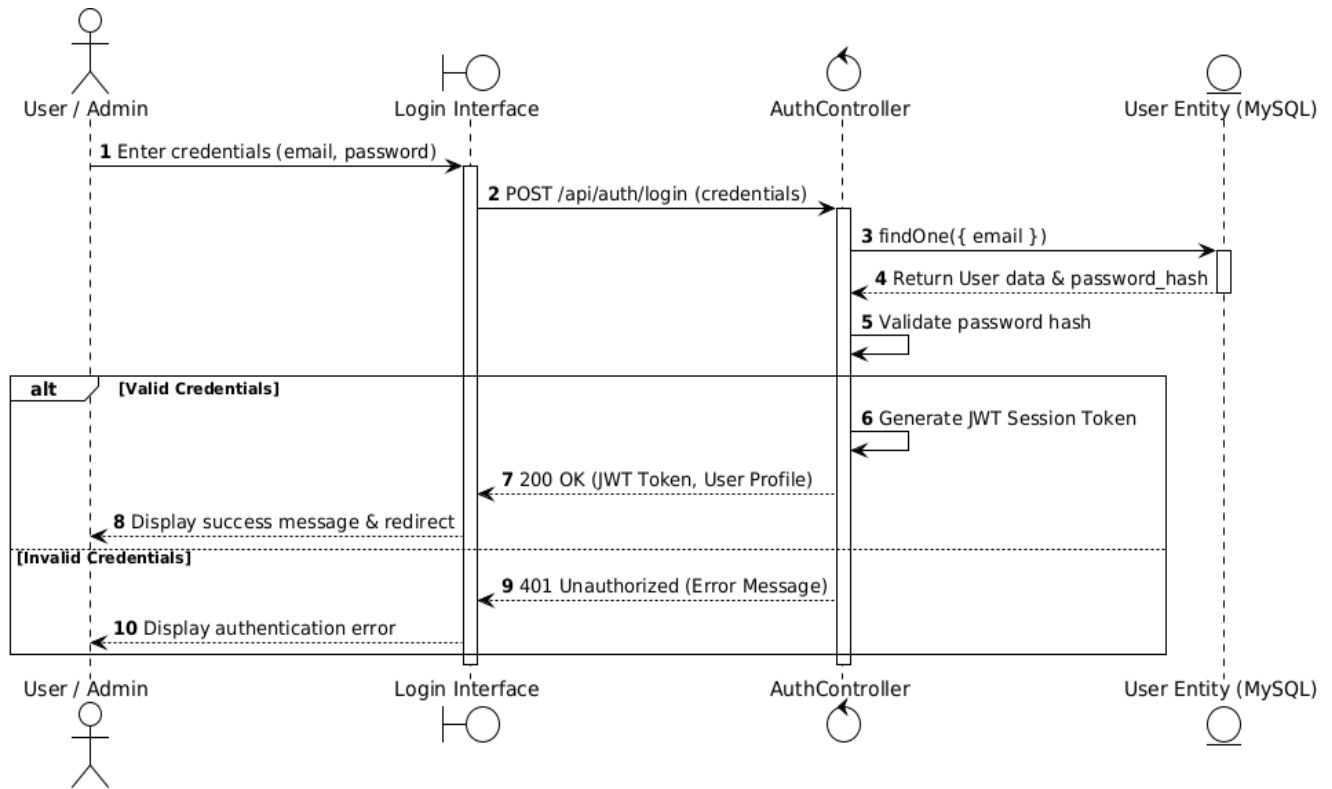


Figure 3.2 Sequence Diagram User Authentication

- Create Telecom Offer Sequence

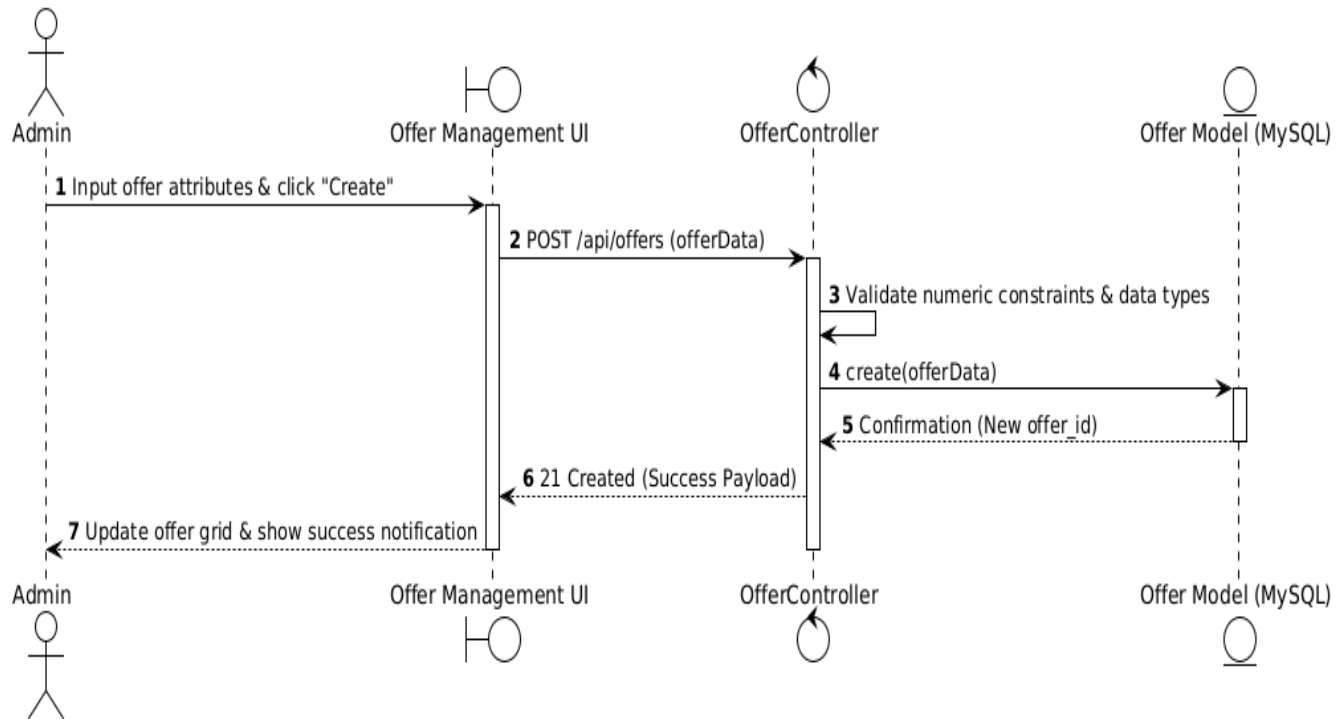


Figure 3.3 Sequence Diagram Create Telecom Offer

- Create Customer Profile Sequence

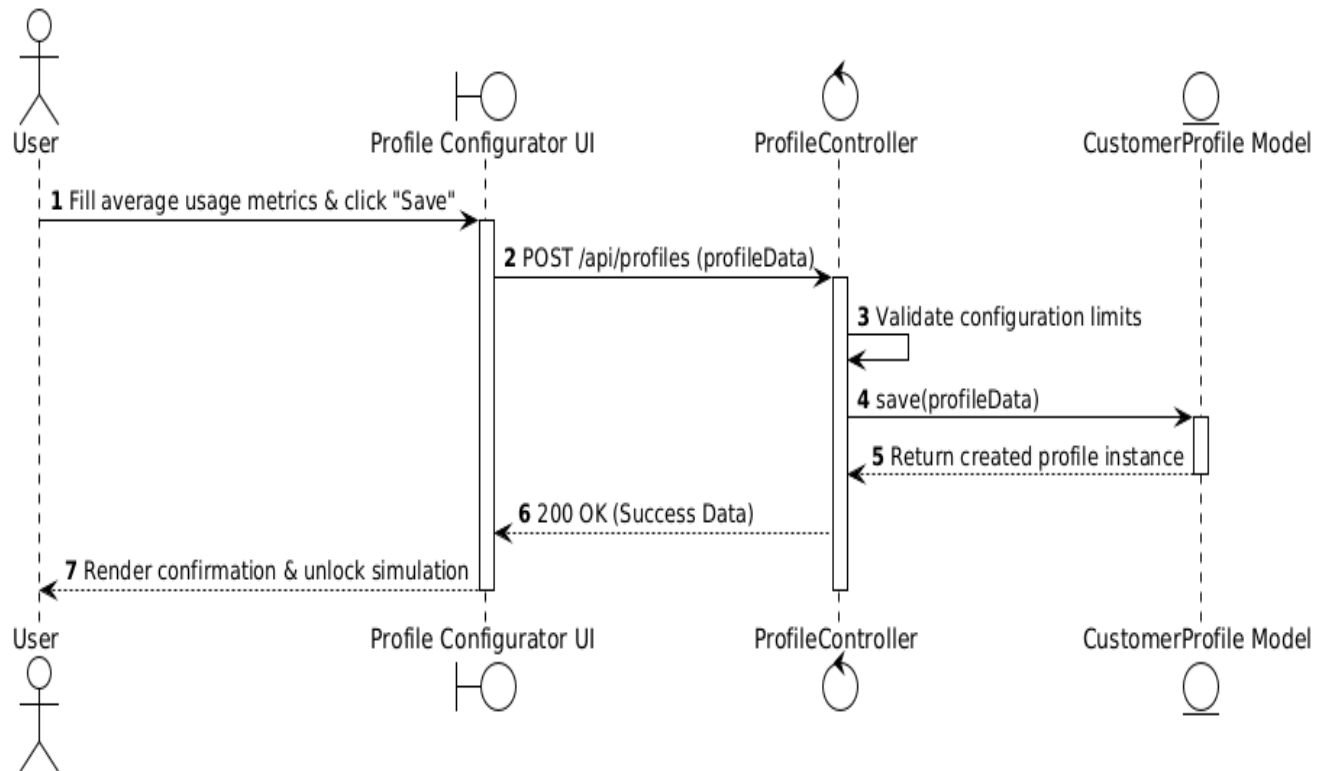


Figure 3.4 Sequence Diagram Create Customer Profile

3.3.3 Sprint 1 Class Diagram

During this sprint, the class diagram was refined to include the entities: User, Offer, Option, OfferOption, and CustomerProfile, with their attributes, methods, and relationships.

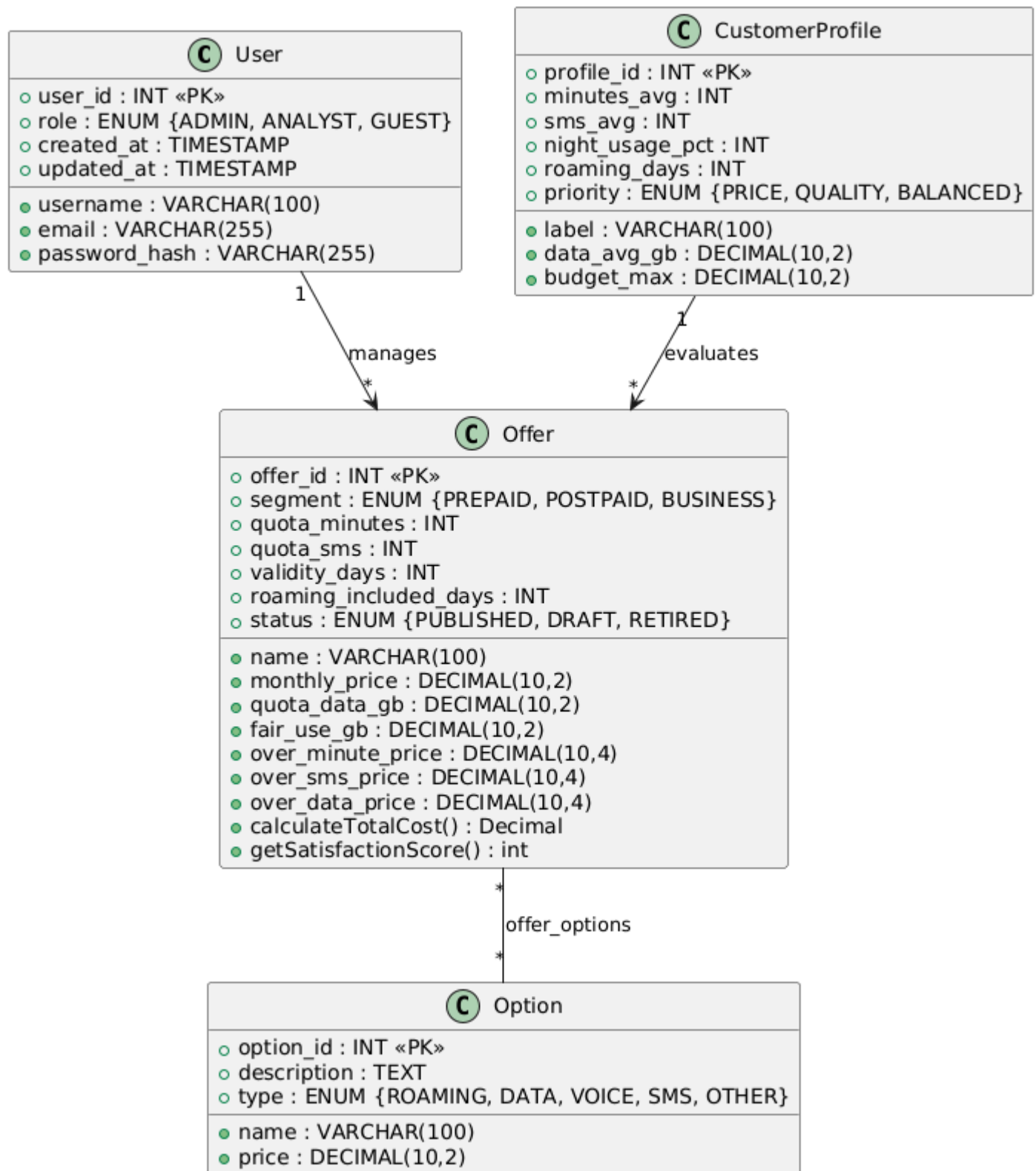


Figure 3.5 Updated Class Diagram

3.3.4 Detailed Use Case Descriptions

In order to better formalize system functionalities, the main use cases of Sprint 1 are described in detail using structured tables.

Use Case 1: User Authentication (Login):

Table 3.2 use case 1 login

Use Case	Login
Actors	Administrator, Analyst, Guest
Pre-condition	The user must have a valid account with correct email and password.
Nominal Scenario	1. The system displays the login interface. 2. The user enters email and password, then submits. 3. The system verifies the credentials against the database. 4. The system generates a JWT token (24h validity). 5. The system redirects the user to the dashboard based on their role.
Post-condition	The user is authenticated and can access the system according to their role.
Alternative/Exception	Invalid credentials: Display error message “Invalid email or password.” Return to step 2.

Use Case 2: Logout:

Table 3.3 use case 2 logout

Use Case	Logout
Actors	All users
Pre-condition	The user must be logged in.
Nominal Scenario	1. The user clicks the logout button. 2. The system invalidates the JWT token. 3. The system redirects the user to the login page.
Post-condition	The user is logged out and the session is terminated.

Use Case 3: Manage Users:

Table 3.4 use case 3 manage users

Use Case	Manage Users
Actors	Administrator
Pre-condition	The administrator must be authenticated with ADMIN role.
Nominal Scenario	1. The admin accesses the user management interface. 2. The system displays the list of existing users. 3. The admin can create a new user (name, email, role, password). 4. The admin can edit an existing user's details. 5. The admin can delete a user. 6. The system validates input and updates the database.
Post-condition	User data is created, updated, or deleted in the database.
Alternative/Exception	Duplicate email: Display error "Email already in use." Invalid input: Display validation error message.

Use Case 4: Manage Telecom Offers:

Table 3.5 use case 4 manage offers

Use Case	Manage Telecom Offers
Actors	Administrator, Analyst
Pre-condition	The user must be authenticated.
Nominal Scenario	1. The user accesses the offers management module. 2. The system displays the list of offers (filterable by status). 3. The user creates a new offer with parameters (name, segment, price, quotas, overage rates, validity, fair-use). 4. The user can edit or delete existing offers. 5. The user can change offer status (Published/Draft/Retired). 6. The system validates and saves changes.
Post-condition	Offer data is stored or updated in the database.
Alternative/Exception	Missing required fields: Display validation error. Unauthorised role: Access denied (403).

Use Case 5: Manage Customer Profiles:

Table 3.6 use case 5 manage customer profiles

Use Case	Manage Customer Profiles
Actors	Administrator, Analyst
Pre-condition	The user must be authenticated.
Nominal Scenario	1. The user accesses the profiles management interface. 2. The system displays existing profiles. 3. The user creates a new profile with usage data (minutes_avg, sms_avg, data_avg_gb, budget_max, priority). 4. The system auto-detects the segment based on usage patterns. 5. The user can edit or delete profiles. 6. The system validates and saves changes.
Post-condition	Profile data is stored in the database.

Alternative/Exception	Invalid input: Display validation error.
-----------------------	--

Use Case 6: Password Reset:

Table 3.7 use case 6 password Reset

Use Case Name	Password Reset
Actor	All users
Description	Allows users to reset their password
Preconditions	User must have a registered email
Postconditions	Password is updated
Main Scenario	1. User requests reset 2. System sends reset link 3. User sets new password
Alternative Scenario	Invalid email → Error message

3.4 Implementation

3.4.1 Backend Implementation

The backend was developed using Node.js and Express.js, following a modular architecture.

Authentication System

- Implementation of secure authentication using JWT (JSON Web Tokens)
- Password hashing using bcrypt
- Role-Based Access Control (RBAC) to manage permissions

User Management

- CRUD operations for users
- Role assignment (Admin, Analyst, Guest)
- Secure API endpoints

Telecom Offers Management

- Creation, modification, and deletion of telecom offers
- Categorization (voice, SMS, data, packs)
- Lifecycle management (Draft, Published, Retired)

Customer Profiles Management

- Creation of usage-based profiles
- Parameters: call duration, SMS usage, data consumption, budget
- Storage and retrieval from MySQL database

Database Design

- Relational schema implemented in MySQL
- Use of foreign keys to ensure data integrity
- Optimized queries for performance

3.4.2 Frontend Implementation

The frontend was developed using React (Vite) and Tailwind CSS, focusing on usability and responsiveness.

Authentication Interface

- Login page with validation
- Secure token storage
- Role-based redirection

Dashboard Interface

- Overview of system modules
- Navigation between features

Management Interfaces

- User management interface (CRUD operations)
- Offers management interface
- Customer profiles management interface

UI/UX Design

- Responsive design for multiple devices
- Clean and modern interface using Tailwind CSS
- Reusable components for scalability

3.5 Realisation

This section presents the graphical interfaces of the developed system, illustrating the main functionalities implemented during Sprint 1.

- Authentication Interface

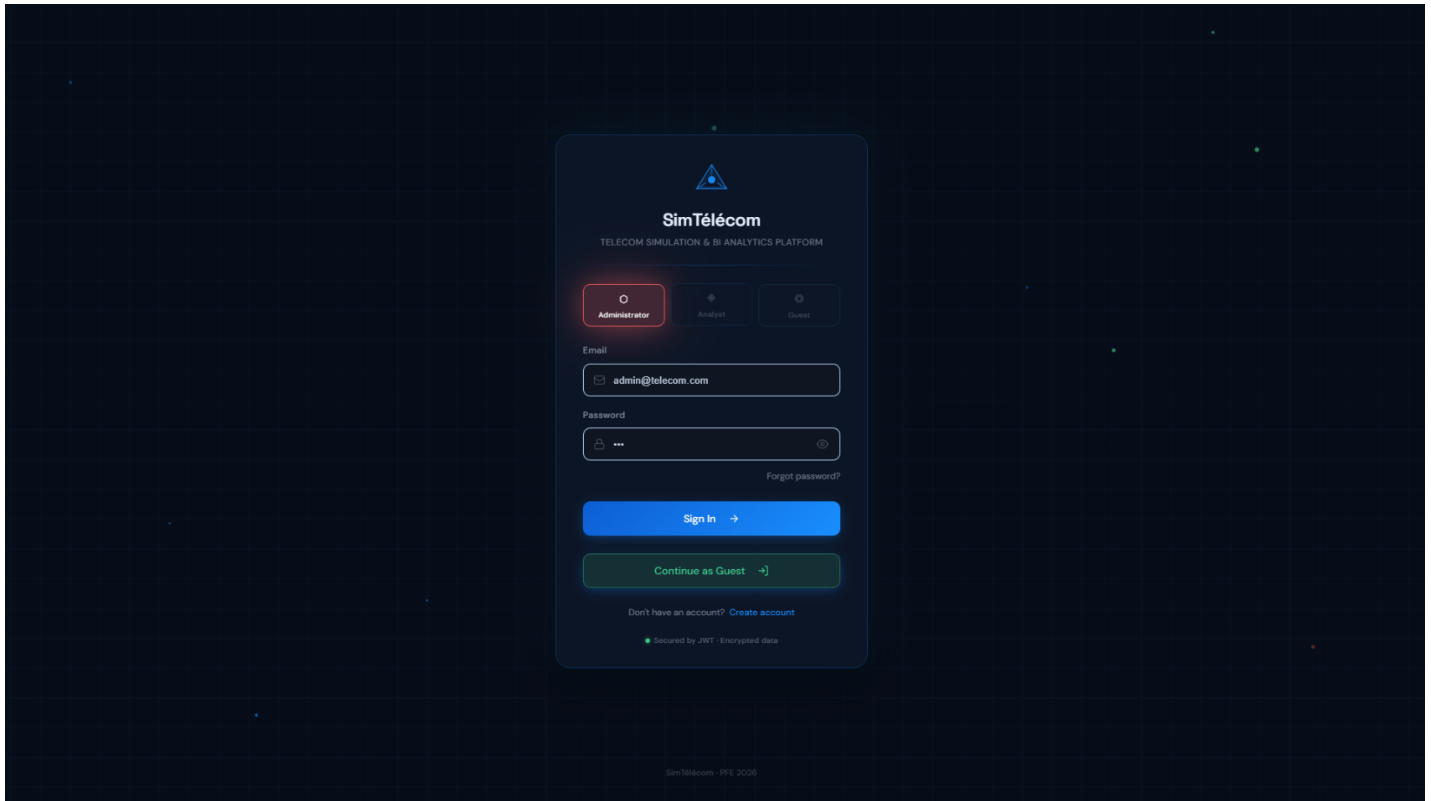


Figure 3.6 Login Interface

- User Management Interface

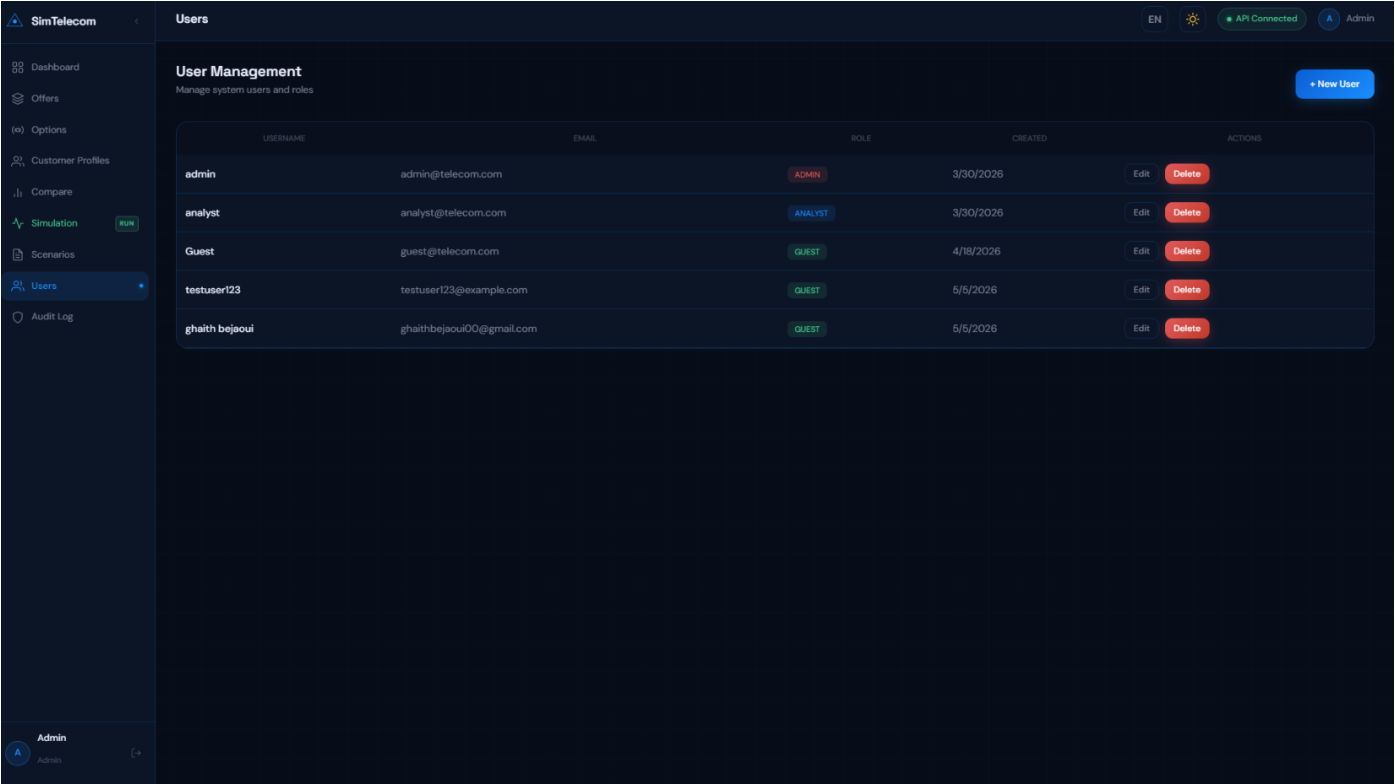


Figure 3.7 User Management

This interface enables administrators to create, update, and delete user accounts.

- Telecom Offers Management Interface

NAME	SEGMENT	PRICE	DATA	MINUTES	SMS	VALIDITY	STATUS	ACTIONS
Offer 1	POSTPAID	37.56 TND	56.82 GB	979	1745	30d	Active	Edit Delete
Offer 2	POSTPAID	34.83 TND	55.07 GB	369	528	30d	Active	Edit Delete
Offer 3	PREPAID	21.55 TND	15.40 GB	257	1281	30d	Active	Edit Delete
Offer 4	POSTPAID	21.75 TND	57.81 GB	838	1183	30d	Active	Edit Delete
Offer 5	POSTPAID	27.59 TND	43.30 GB	1033	433	30d	Draft	Edit Delete
Offer 6	PREPAID	21.65 TND	8.36 GB	220	898	30d	Active	Edit Delete
Offer 7	PREPAID	20.74 TND	14.40 GB	635	1346	30d	Active	Edit Delete
Offer 8	POSTPAID	38.84 TND	63.55 GB	1605	1739	30d	Active	Edit Delete
Offer 9	PREPAID	13.73 TND	8.76 GB	787	468	30d	Active	Edit Delete
Offer 10	POSTPAID	42.28 TND	56.50 GB	1736	1109	30d	Active	Edit Delete
Offer 11	PREPAID	19.45 TND	4.06 GB	572	950	30d	Active	Edit Delete
Offer 12	PREPAID	17.61 TND	2.67 GB	425	1234	30d	Active	Edit Delete
Offer 13	PREPAID	24.36 TND	14.62 GB	347	1968	30d	Active	Edit Delete
Offer 14	PREPAID	11.12 TND	4.36 GB	387	1626	30d	Active	Edit Delete

Figure 3.8 Offers Management

This interface allows the management of telecom offers, including creation, modification, and deletion.

- Customer Profiles Interface

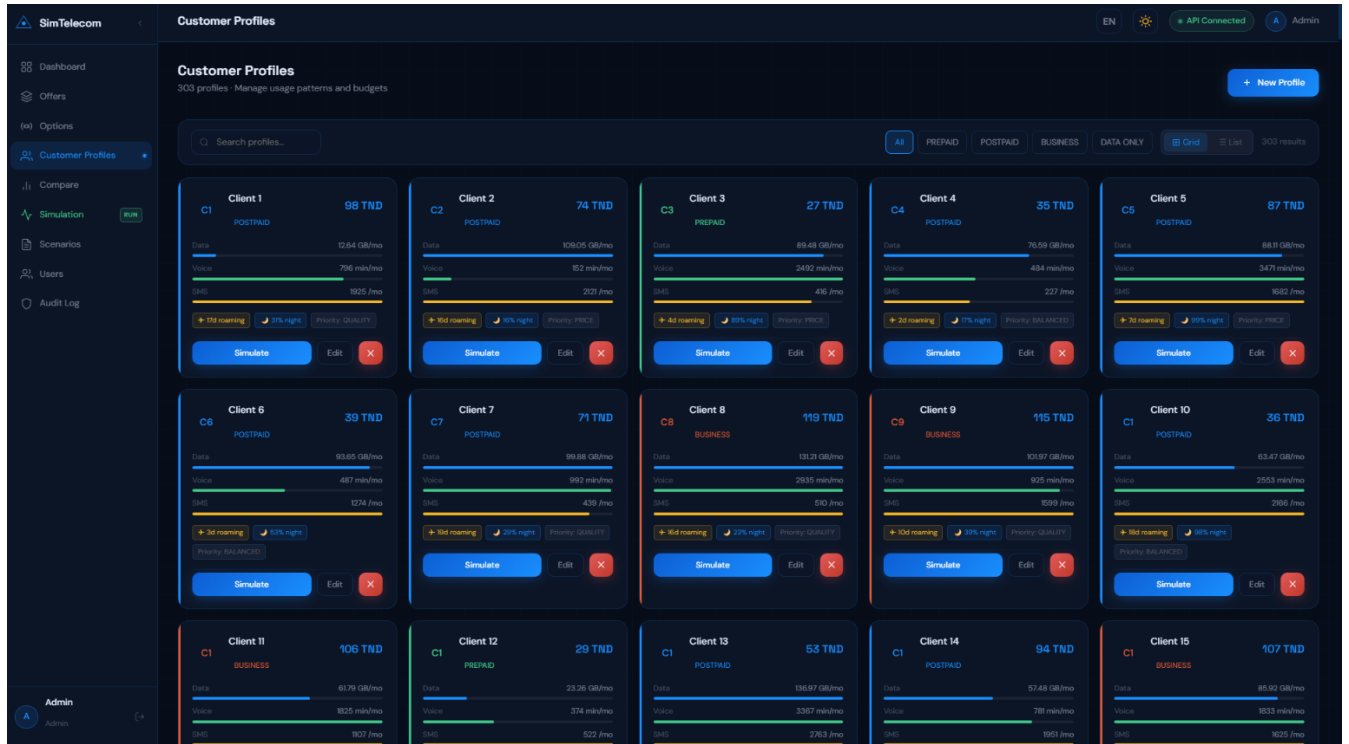


Figure 3.9 Customer Profiles

This interface allows the creation and management of customer usage profiles.

3.6 Testing and Validation

Testing was carried out to verify the proper functioning of the implemented features.

Table 3.8 testing validation

Test Case	Description	Expected Result	Actual Result	Status
Login Test	User logs in with valid credentials	Access granted, JWT issued	Access granted	Passed
Login Fail	Invalid credentials submitted	Error message displayed	Error displayed	Passed
Create User	Admin creates new user	User saved in database	User saved	Passed

Create Offer	Create telecom offer with all parameters	Offer stored with correct status	Offer stored	Passed
Create Profile	Create customer profile with usage data	Profile saved with auto-segment	Profile saved	Passed
Role Access	Guest tries to access admin features	403 Forbidden	403 returned	Passed
Offer Filter	Filter offers by PUBLISHED status	Only published offers shown	Correct filter	Passed

All test cases were successfully validated, confirming the reliability of the system.

Conclusion

Sprint 1 successfully established the core foundation of the platform by implementing essential management modules and system architecture. The backend and frontend components are now fully operational, providing a solid base for integrating advanced features such as simulation and business intelligence in the upcoming sprints.

CHAPTER 4: Sprint 2 – Simulation Engine and Offer Comparison

Summary

Introduction	46
4.1 Sprint 2 Objectives	46
4.2 Sprint 2 Backlog	46
4.3 System Design	47
4.3.1 Use Case Diagram	47
4.3.2 Sequence Diagrams.....	48
4.3.3 Activity Diagram	49
4.3.4 Detailed Use Case Descriptions	50
4.4 Implementation.....	53
4.4.1 Simulation Engine Logic	53
4.4.2 Backend Implementation.....	53
4.4.3 Frontend Implementation.....	53
4.5 Realisation	54
4.6 Testing and Validation	59
4.7 Conclusion	59

Introduction

Sprint 2 focuses on the development of the core functionality of the platform: the Simulation Engine. This component enables users to evaluate telecom offers based on customer usage profiles and compare multiple offers to identify the most suitable one.

The objective of this sprint is to transform the system from a simple management platform into a decision-support tool by implementing advanced simulation and comparison features.

4.1 Sprint 2 Objectives

The main objectives of this sprint are:

- Develop the simulation engine logic
- Implement mono-offer simulation
- Implement multi-offer comparison
- Design recommendation mechanism
- Integrate customer profiles into simulation process
- Display simulation results in a clear and structured format

4.2 Sprint 2 Backlog

Table 4.1 sprint 2 backlog

ID	User Story	Task Description	Type	Priority
6.1	Mono-Offer Simulation	Implement single simulation API endpoint	Back-End	High
6.2	Mono-Offer Simulation	Build simulation form and results display	Front-End	High
7.1	Multi-Offer Comparison	Implement comparison API with ranking	Back-End	High
7.2	Multi-Offer Comparison	Build comparison interface	Front-End	High
8.1	Recommendation	Implement recommendation algorithm (top N)	Back-End	High

9.1	Batch Simulation	Implement one-offer-vs-many-profiles batch	Back-End	High
9.2	Batch Simulation	Build batch results interface with summary	Front-End	High

4.3 System Design

4.3.1 Use Case Diagram

This diagram represents the interactions between users and the simulation functionalities.

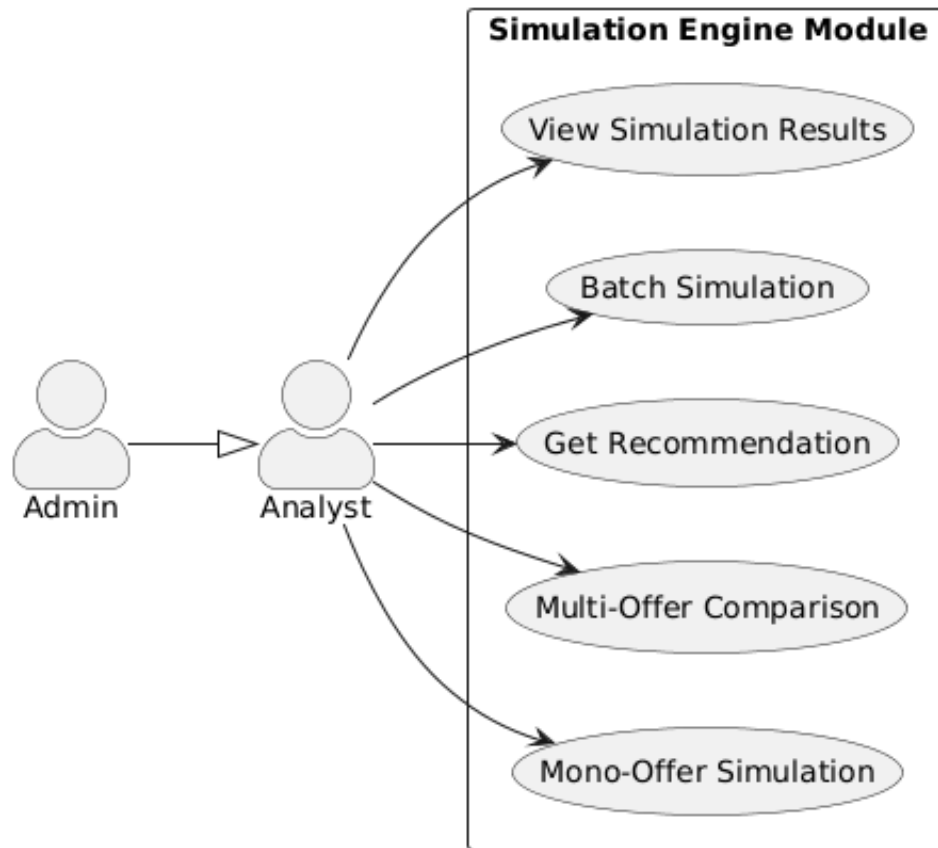


Figure 4.1 Sprint 2 Use Case Diagram

4.3.2 Sequence Diagrams

- Mono-offer Simulation Sequence

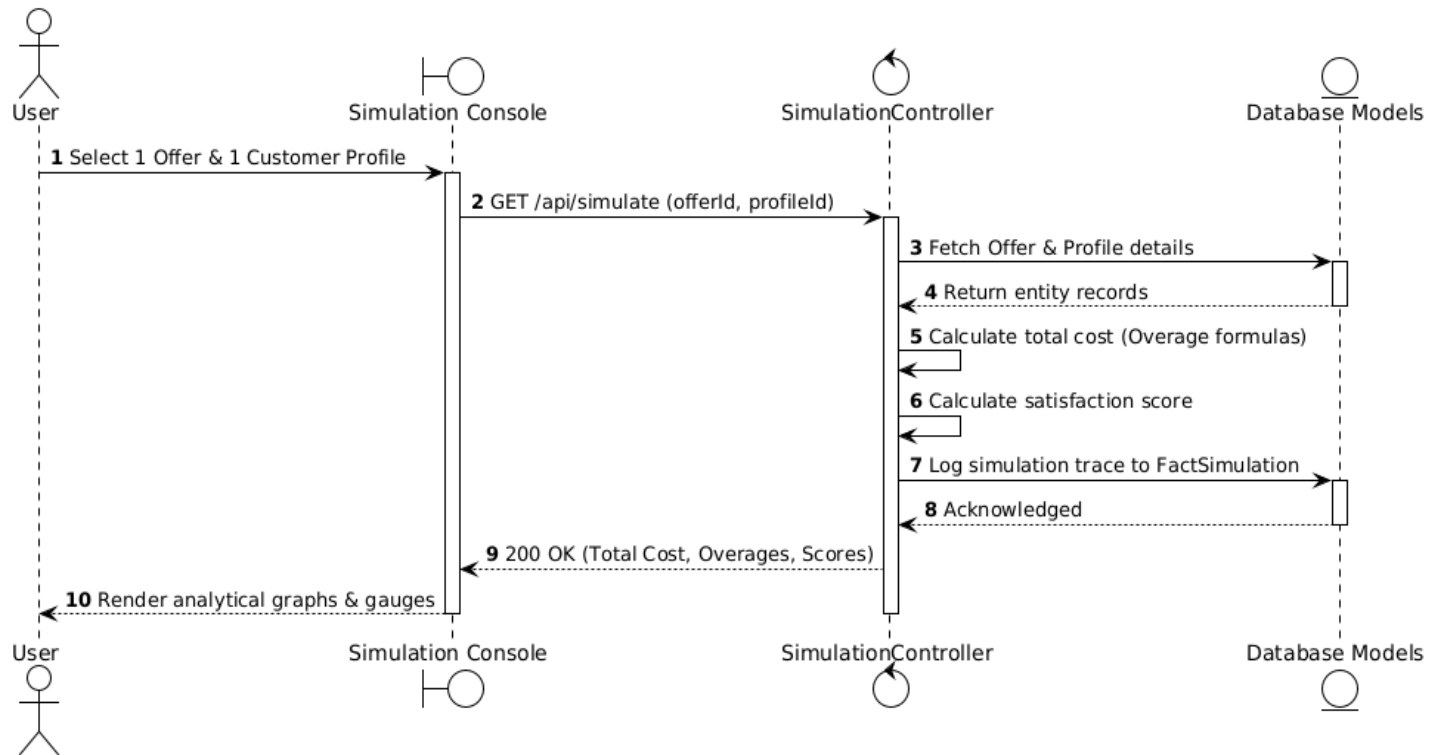


Figure 4.2 Mono-Offer Simulation Sequence Diagram

- Multi-offer Comparison Sequence

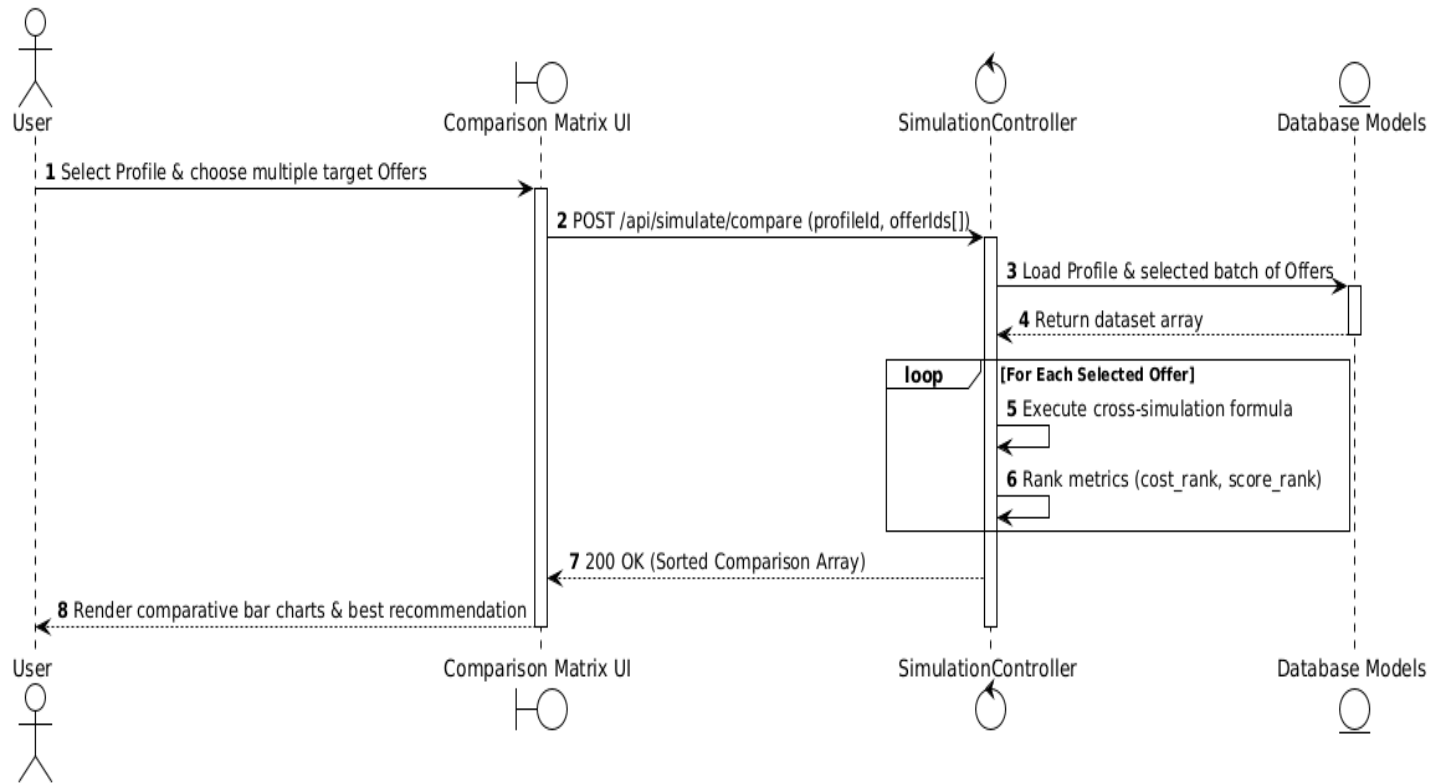


Figure 4.3 Multi-Offer Comparison Sequence Diagram

4.3.3 Activity Diagram

The activity diagram describes the workflow of the simulation process, from selecting a customer profile to generating results.

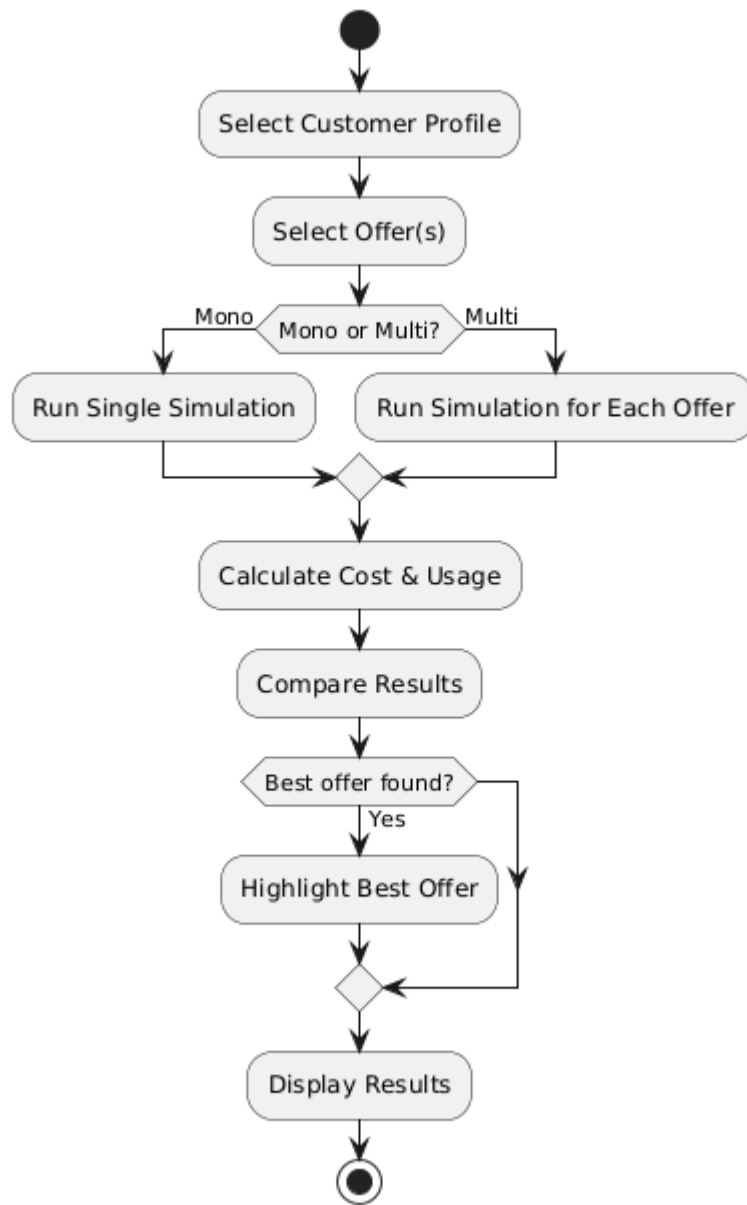


Figure 4.4 Simulation Activity Diagram

4.3.4 Detailed Use Case Descriptions

Use Case 1: Mono-Offer Simulation :

Table 4.2 use case mono-offer simulation

Use Case	Mono-Offer Simulation
Actors	Analyst, Administrator
Pre-condition	At least one customer profile and one published offer must exist in the system.
Nominal Scenario	1. The user selects the “Single” simulation mode. 2. The user selects a customer profile (or enters custom parameters). 3. The user selects a telecom offer. 4. The user clicks “Run Simulation.” 5. The system calculates: total cost, overage charges (minutes, SMS, data), and satisfaction score. 6. The system displays the detailed cost breakdown and satisfaction score.
Post-condition	Simulation result is generated and displayed. Result is logged in fact_simulations table.
Alternative/Exception	Missing profile or offer: Display error “Please select both a profile and an offer.”

Use Case 2: Multi-Offer Comparison :

Table 4.3 use case 2 multi-offer comparison

Use Case	Multi-Offer Comparison
Actors	Analyst, Administrator
Pre-condition	At least one profile and multiple published offers must exist.
Nominal Scenario	1. The user selects the “Compare” mode. 2. The user selects a customer profile. 3. The user selects multiple offers (checkboxes). 4. The user clicks “Compare.” 5. The system evaluates each offer and ranks them by cost and satisfaction. 6. Results are displayed side-by-side with ranking indicators.
Post-condition	Comparison results displayed with cost_rank and score_rank.
Alternative/Exception	Fewer than 2 offers selected: Display error message.

Use Case 3: Intelligent Recommendation:

Table 4.4 use case 3 Recommendation system

Use Case	Intelligent Recommendation
Actors	Analyst, Administrator
Pre-condition	Customer profile parameters must be defined.
Nominal Scenario	1. The user selects the “Recommend” mode. 2. The user enters profile parameters or selects an existing profile. 3. The user optionally filters by segment. 4. The user clicks “Get Recommendations.” 5. The system evaluates all published offers and returns the top 5 sorted by satisfaction/cost ratio. 6. Each recommendation includes justification text.
Post-condition	Top recommended offers are displayed with explanations.

4.4 Implementation

4.4.1 Simulation Engine Logic

The simulation engine implements the following calculation formulas:

$$\text{Total Cost} = \text{Monthly Price} + \text{Overage Minutes Cost} + \text{Overage SMS Cost} + \text{Overage Data Cost} + \text{Roaming Cost} - \text{Options Credits}$$

Where overage costs are calculated as: $\max(0, \text{usage} - \text{quota}) \times \text{overage rate}$, for each dimension (minutes, SMS, data GB).

Satisfaction Score (0–100):

- Base score = 100
- +10 if $\text{total_cost} / \text{budget} \leq 0.7$ (budget-friendly)
- -30 if $\text{total_cost} / \text{budget} > 1.0$ (over budget)
- +10 for BUSINESS segment, +5 for POSTPAID
- +2 per linked option (capped at +10)
- -20 if data usage exceeds fair-use threshold
- -10 if any overage exists
- Score is bounded to $[0, 100]$

All monetary values are rounded to 2 decimal places for precision.

4.4.2 Backend Implementation

The simulation module comprises 628 lines of route code with four API endpoints:

- POST /api/simulation – Single offer simulation
- POST /api/simulation/compare – Multi-offer comparison with ranking
- POST /api/simulation/recommend – Top N recommendations
- POST /api/simulation/batch – One offer vs. multiple profiles

4.4.3 Frontend Implementation

The simulation interface provides four modes accessible via tabs: Single, Compare, Recommend, and Batch. Results include detailed cost breakdowns, satisfaction scores, and visual ranking indicators.

4.5 Realisation

This section presents the main interfaces developed during Sprint 2, focusing on the simulation engine and offer comparison functionalities.

- Simulation Dashboard Interface

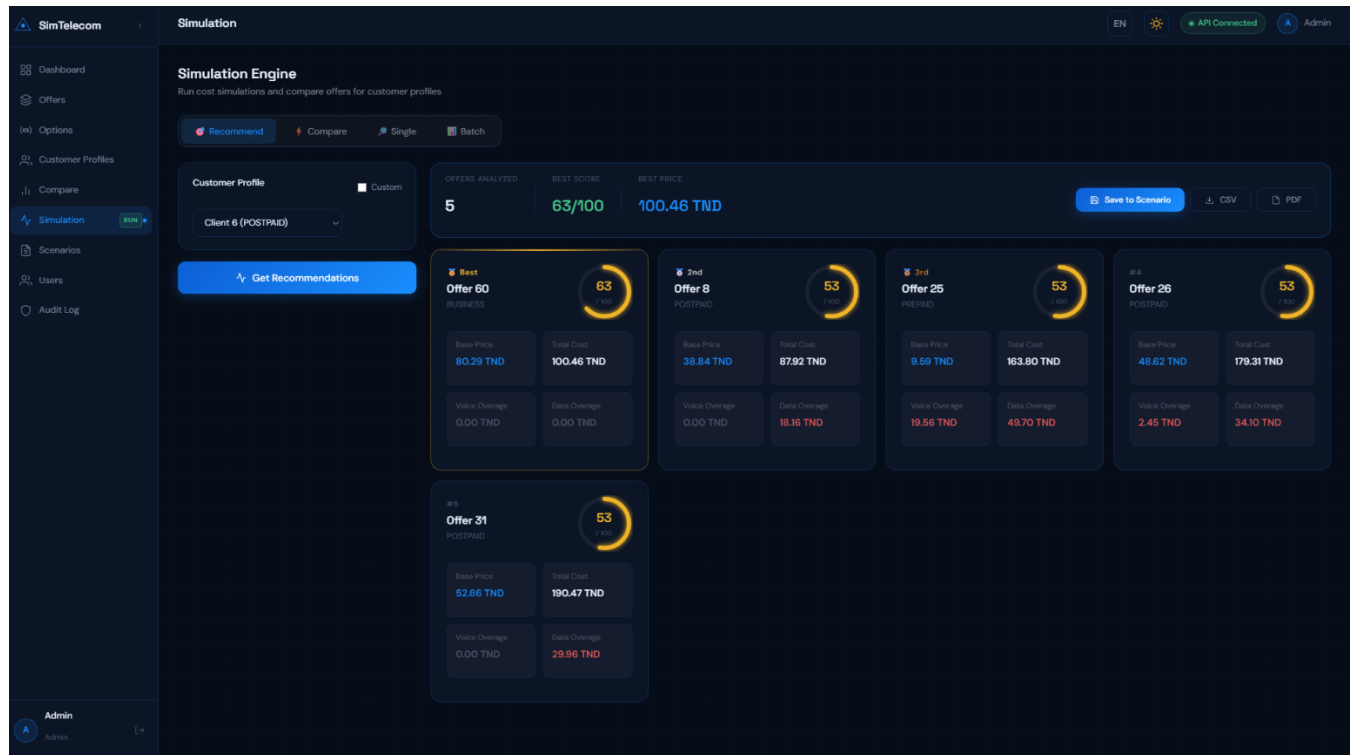


Figure 4.5 Simulation Main Page

This interface represents the main simulation dashboard, where users can choose between different simulation modes such as single simulation, comparison, recommendation, and batch processing.

- Single Offer Simulation

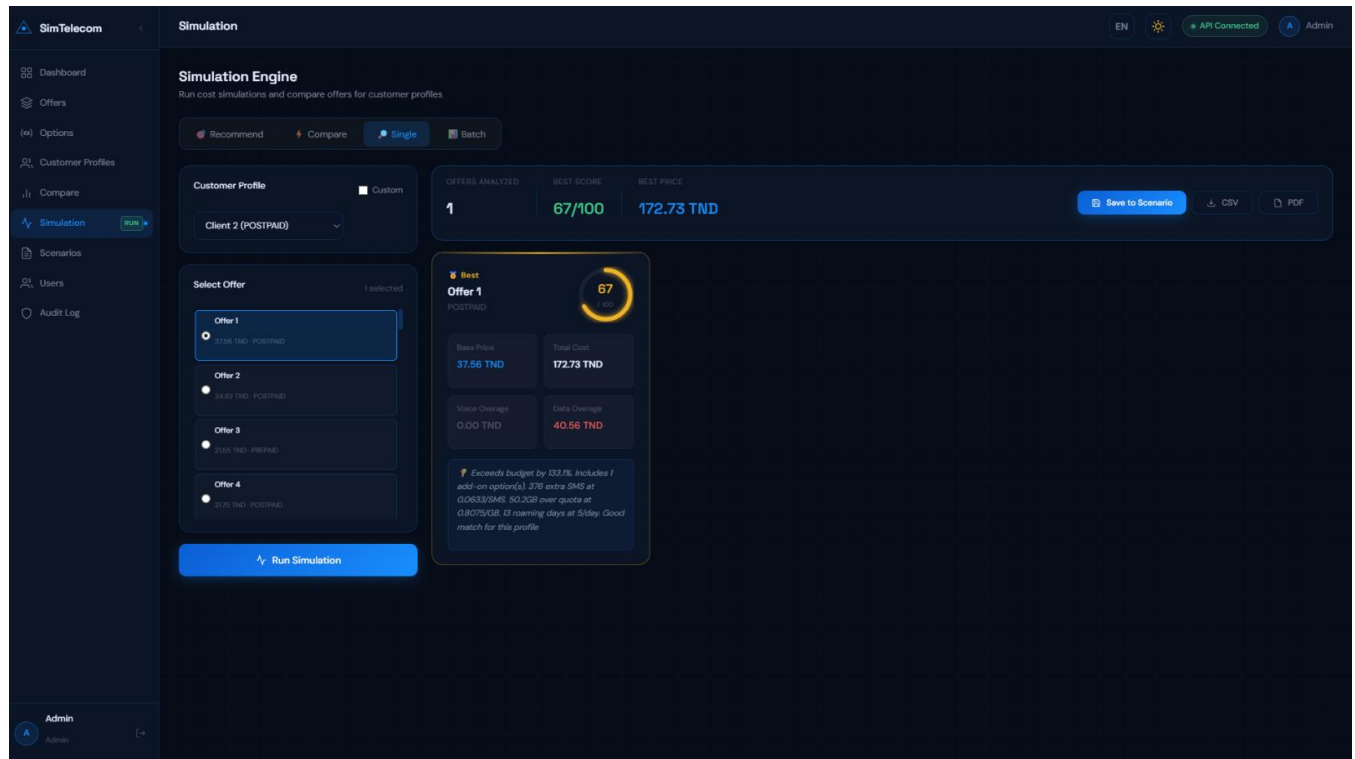


Figure 4.6 Single Simulation

This interface allows the user to evaluate a single telecom offer based on a selected customer profile. It displays detailed results including cost, usage, and remaining balance.

- Multi-Offer Comparison

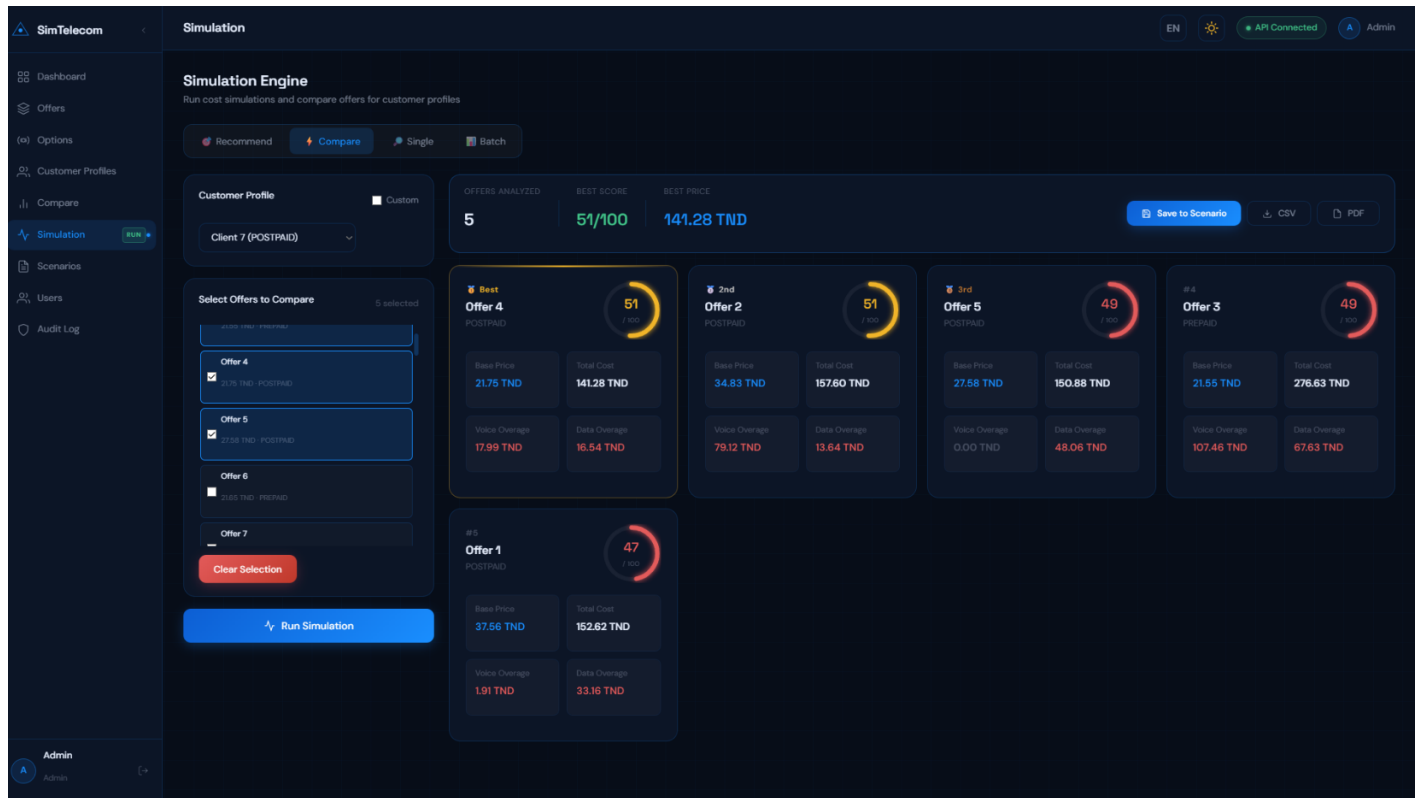


Figure 4.7 Comparison Simulation

This interface enables the comparison of multiple telecom offers simultaneously. It highlights differences in pricing, consumption, and overall performance, allowing users to easily identify the most suitable option.

- Recommendation Mode

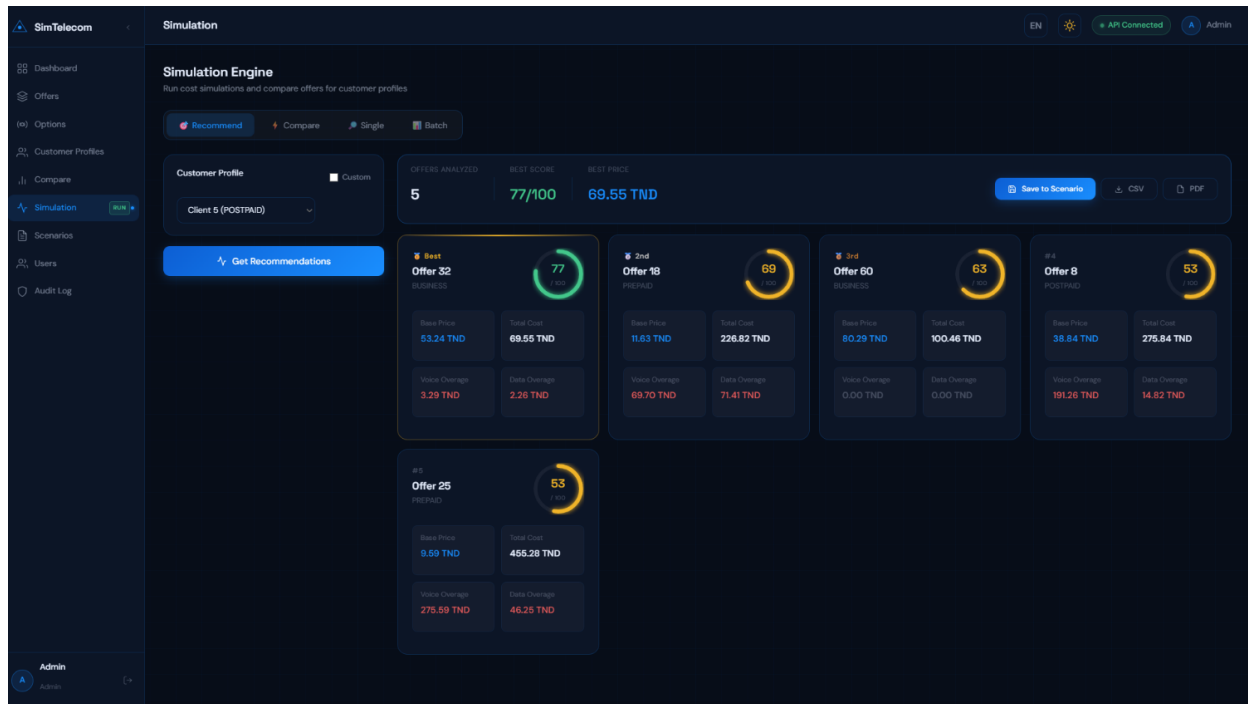


Figure 4.8 Recommendation

This interface automatically suggests the best telecom offer based on the selected customer profile and simulation results, helping users make optimal decisions.

- Batch Simulation

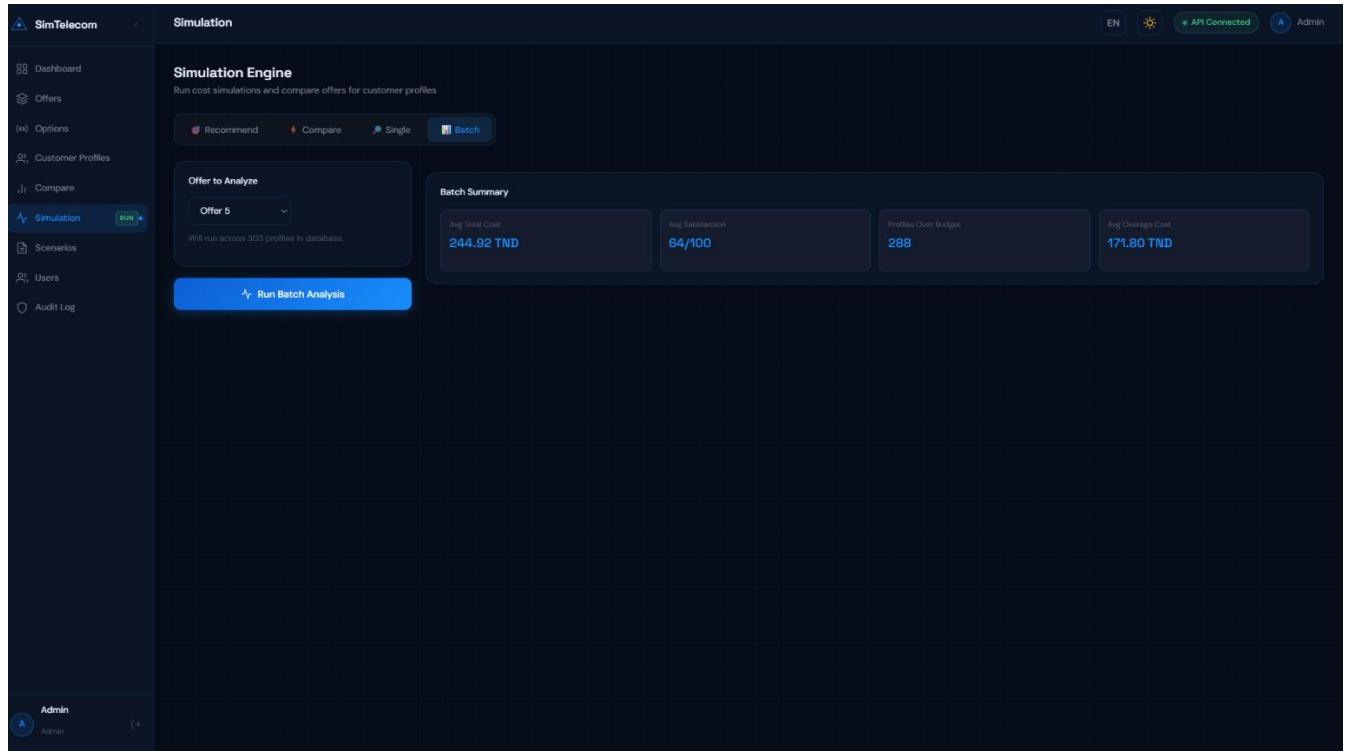


Figure 4.9 Batch Simulation

This interface enables the execution of simulations for a single telecom offer across multiple customer profiles simultaneously. It provides a global analysis of the offer's performance over a large dataset.

The system generates a Batch Summary, which includes:

- **Average Total Cost:** The mean cost of the offer across all evaluated profiles
- **Average Satisfaction Score:** A global satisfaction indicator reflecting how well the offer meets customer needs
- **Profiles Over Budget:** The number of profiles for which the offer exceeds the defined budget
- **Average Overage Cost:** The average extra cost incurred when usage exceeds the offer limits

This functionality allows decision-makers to assess the overall efficiency and suitability of a telecom offer at scale.

4.6 Testing and Validation

Testing was conducted to verify the accuracy and performance of the simulation engine and comparison functionalities.

Table 4.5 testing and validation

Test Case	Description	Expected Result	Actual Result	Status
Mono Simulation	Simulate one offer against a profile	Correct cost and satisfaction	Correct result	Passed
Multi Comparison	Compare 3 offers side-by-side	Accurate ranking	Accurate output	Passed
Recommendation	Get top 5 recommendations	Best offers identified	Correct recommendations	Passed
Batch Simulation	One offer vs. 50 profiles	Summary statistics generated	Correct batch results	Passed
Edge Case	Profile with zero usage	Cost equals base price only	Correct calculation	Passed

All simulation functionalities were successfully tested, ensuring accurate calculations and reliable performance.

Conclusion

Sprint 2 successfully introduced the simulation engine, transforming the platform into a decision-support system. The ability to evaluate and compare telecom offers using four distinct modes provides significant value for marketing and product teams.

CHAPTER 5:Sprint 3 – Scenario Management and Data Export

Summary

Introduction	61
5.1 Sprint 3 Objectives	61
5.2 Sprint 3 Backlog	61
5.3 System Design	62
5.3.1 Use Case Diagram	62
5.3.2 Sequence Diagrams.....	63
5.3.3 Detailed Use Case Descriptions	64
5.4 Implementation.....	66
5.4.1 Scenario Management Module.....	66
5.4.2 Export Functionality.....	67
5.4.3 Backend Implementation.....	67
5.4.4 Frontend Implementation.....	67
5.5 Realisation	67
5.6 Testing and Validation	72
Conclusion.....	73

Introduction

Sprint 3 focuses on enhancing the usability and analytical capabilities of the platform by introducing scenario management and data export functionalities.

This phase allows users to save simulation results, reuse them, and export data for further analysis. These features are essential in transforming the platform into a practical decision-support tool that can be used in real business environments.

5.1 Sprint 3 Objectives

The objectives of this sprint are:

- Implement scenario saving, editing, deletion, and duplication
- Enable export of simulation results in CSV and XLSX formats
- Implement audit logging for all system actions
- Prepare data structure for Business Intelligence integration

5.2 Sprint 3 Backlog

Table 5.1 Sprint 3 backlog

ID	User Story	Task Description	Type	Priority
10.1	Manage Scenarios	Create scenario CRUD API (637 lines)	Back-End	High
10.2	Manage Scenarios	Build scenario management interface	Front-End	High
10.3	Duplicate Scenario	Implement scenario duplication endpoint	Back-End	Medium
11.1	Export Results	Implement CSV/XLSX export (246 lines)	Back-End	Medium
11.2	Export Results	Add export buttons to interfaces	Front-End	Medium

13.1	Audit Logging	Implement audit trail (305 lines)	Back-End	Medium
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5.3 System Design

5.3.1 Use Case Diagram

This diagram illustrates interactions related to scenario management and data export.

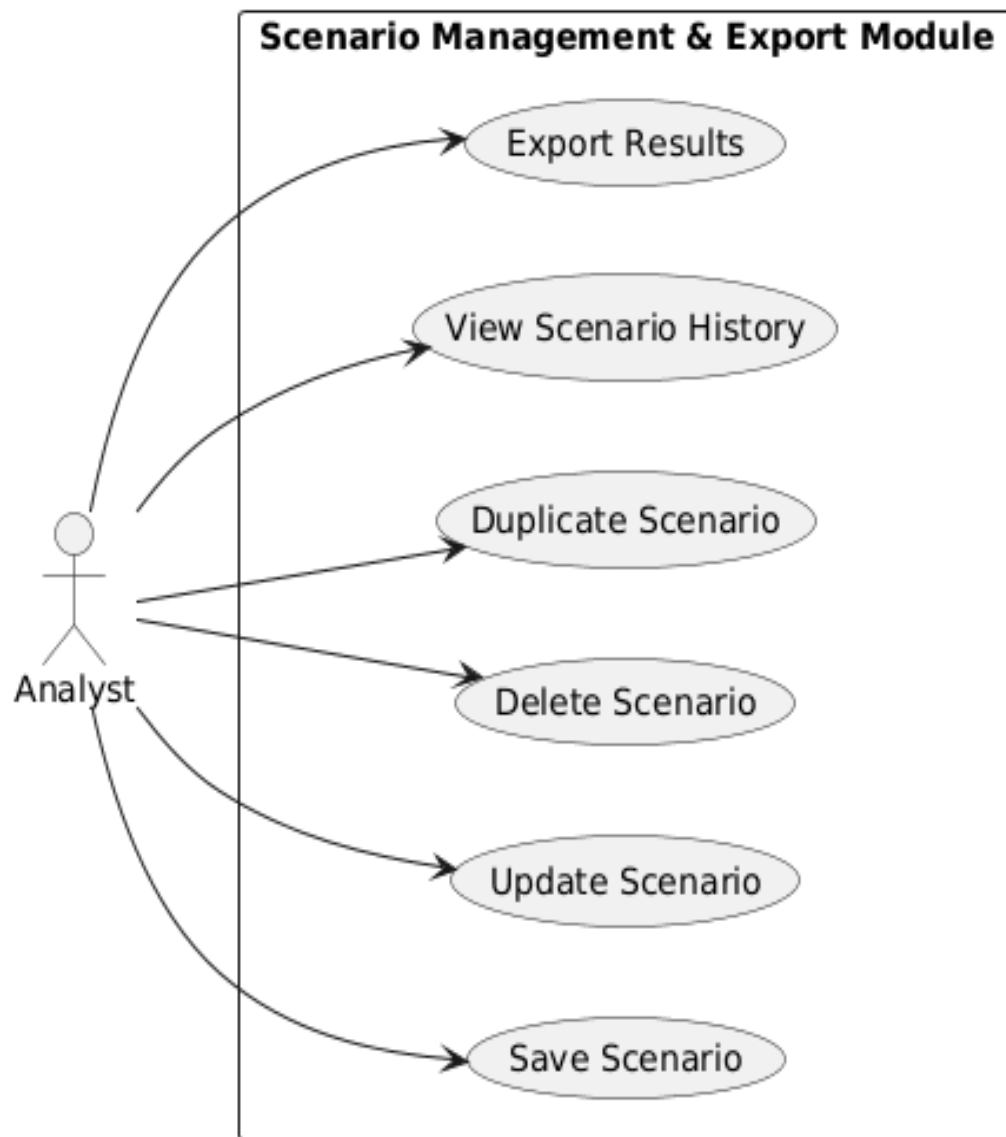
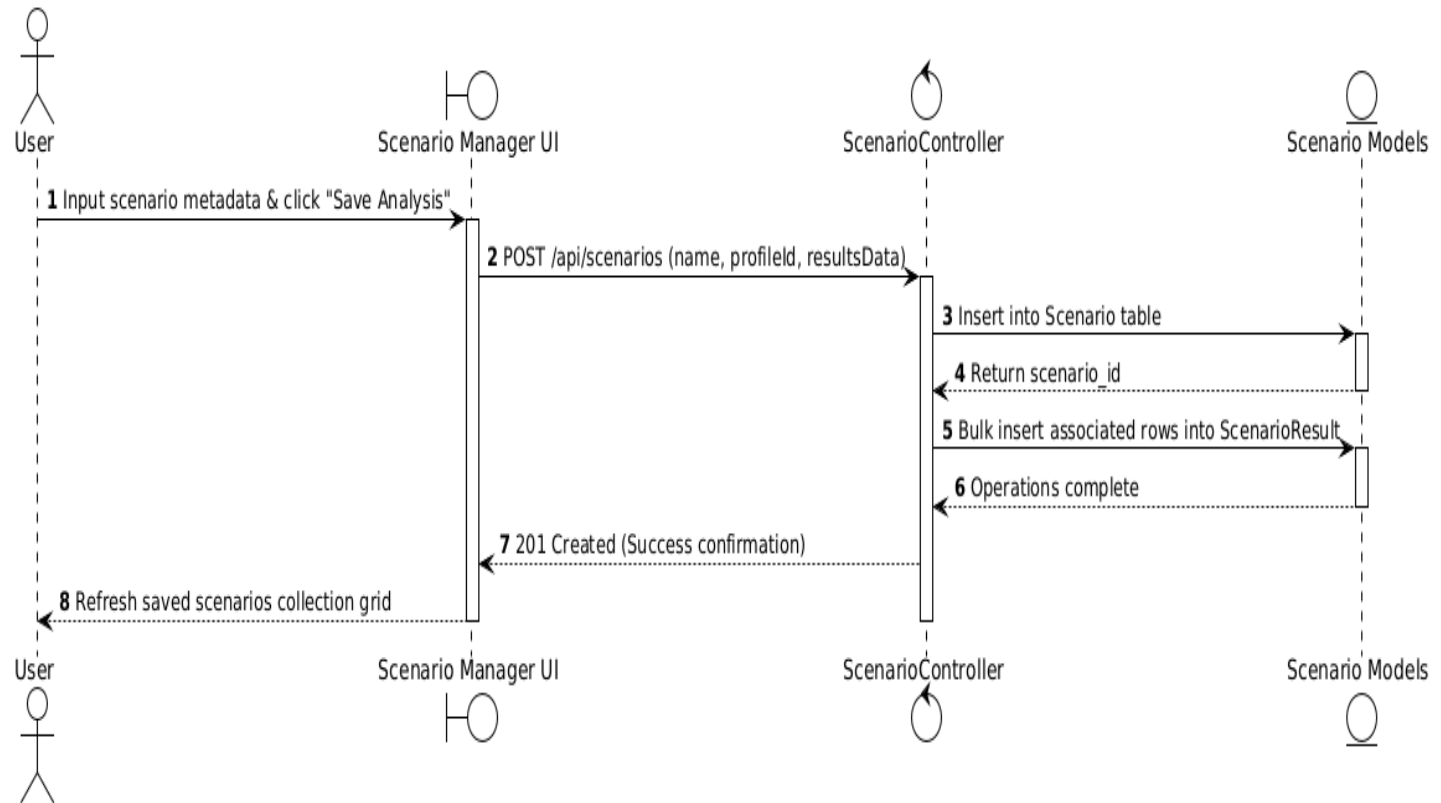


Figure 5.1 Sprint 3 Use Case Diagram

5.3.2 Sequence Diagrams

- Save Scenario Sequence

*Figure 5.2 Save Scenario Sequence Diagram*

- Export Results Sequence

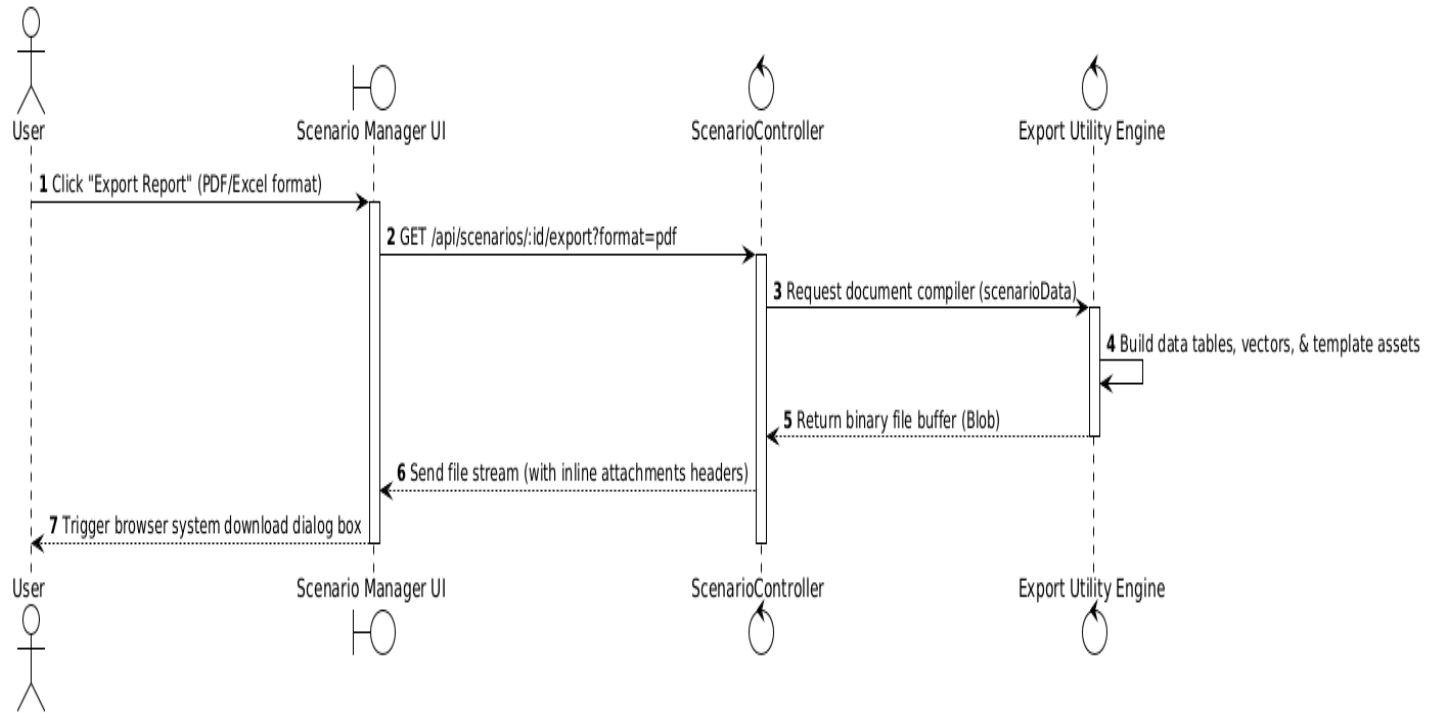


Figure 5.3 Export Results Sequence Diagram

5.3.3 Detailed Use Case Descriptions

Use Case 1: Save Scenario:

Table 5.2 use case 1 save scenario

Use Case	Save Scenario
Actors	Analyst, Administrator
Pre-condition	A simulation must have been completed.
Nominal Scenario	1. The user runs a simulation (single or compare mode). 2. The user clicks “Save to Scenario.” 3. The system displays a form for scenario name and description. 4. The user enters details and submits. 5. The system creates the scenario in DRAFT status and stores associated offer IDs and profile. 6. The system displays a success message.
Post-condition	Scenario is stored in the database with all associated data.
Alternative/Exception	Missing name: Display validation error.

Use Case 2: Export Results:

Table 5.3 use case 2 Export Results

Use Case	Export Results
Actors	Analyst, Administrator
Pre-condition	Simulation results or a saved scenario with results must exist.
Nominal Scenario	1. The user navigates to the scenario details or simulation results. 2. The user clicks “Export CSV” or “Export XLSX.” 3. The system generates the file with all result data. 4. The browser downloads the file automatically.
Post-condition	File is downloaded to the user’s machine.

Use Case 3: Duplicate Scenario:

Table 5.4 use case 3 Duplicate scenario

Use Case	Duplicate Scenario
Actors	Analyst, Administrator
Pre-condition	A scenario must exist in the system.
Nominal Scenario	1. The user selects a scenario from the list. 2. The user clicks “Duplicate.” 3. The system creates a copy with “(Copy)” appended to the name. 4. The new scenario appears in the list with DRAFT status.
Post-condition	A new independent copy of the scenario is created.

5.4 Implementation

5.4.1 Scenario Management Module

This module allows users to manage simulation scenarios efficiently.

Features

- Save simulation results
- Edit existing scenarios
- Delete scenarios
- Duplicate scenarios
- View scenario history

Database Integration

- Scenarios stored with:
 - Profile ID
 - Selected offers

- Simulation results
- Timestamp

5.4.2 Export Functionality

The export module allows users to extract simulation results for external use.

Supported Formats

- CSV
- Excel

Exported Data

- Customer profile details
- Selected offers
- Simulation results
- Cost analysis

5.4.3 Backend Implementation

- API endpoints:
 - /scenarios (CRUD)
 - /export
- Data processing and formatting
- File generation logic

5.4.4 Frontend Implementation

- Scenario management interface
- Export button and options
- User-friendly interaction

5.5 Realisation

This section presents the main interfaces developed during Sprint 3, focusing on scenario management and data export functionalities.

- Scenario Saving Interface

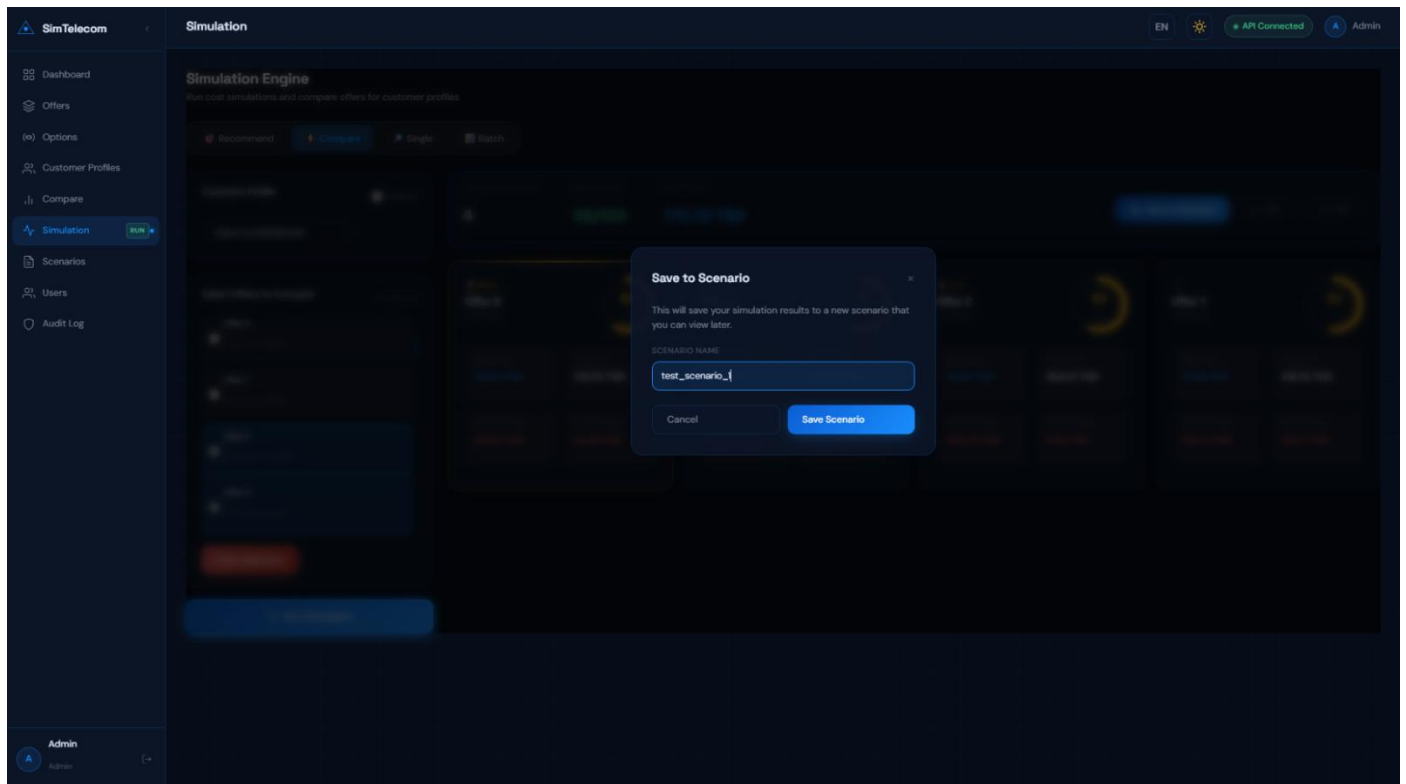


Figure 5.4 Save Scenario

This interface allows the user to save the results of a simulation as a scenario. The system stores all relevant data, including the selected customer profile, telecom offers, and simulation results.

- Scenario Management Interface

NAME	STATUS	UPDATED	ACTIONS
test_scenario_1 Simulation: compare mode	Active	5/5/2026	View Edit Run & Save Copy Delete
taz Simulation: recommend mode	Active	5/4/2026	View Edit Run & Save Copy Delete
test scenario test	Draft	5/4/2026	View Edit Run & Save Copy Delete
API Flow	Active	4/18/2026	View Edit Run & Save Copy Delete
API Flow	Active	4/18/2026	View Edit Run & Save Copy Delete
Test Scenario from Sim	Active	4/18/2026	View Edit Run & Save Copy Delete
Test Save Scenario	Active	4/18/2026	View Edit Run & Save Copy Delete
API Flow	Active	4/18/2026	View Edit Run & Save Copy Delete
Final Test Scenario	Active	4/18/2026	View Edit Run & Save Copy Delete
client 2 offer 1,2 Simulation: compare mode	Active	4/18/2026	View Edit Run & Save Copy Delete
21 Simulation: compare mode	Active	4/18/2026	View Edit Run & Save Copy Delete
API Flow	Active	4/18/2026	View Edit Run & Save Copy Delete

Figure 5.5 Scenario List

This interface displays all saved scenarios. Users can view, update, delete, or duplicate existing scenarios, enabling efficient management and reuse of simulation data.

- Scenario Details View

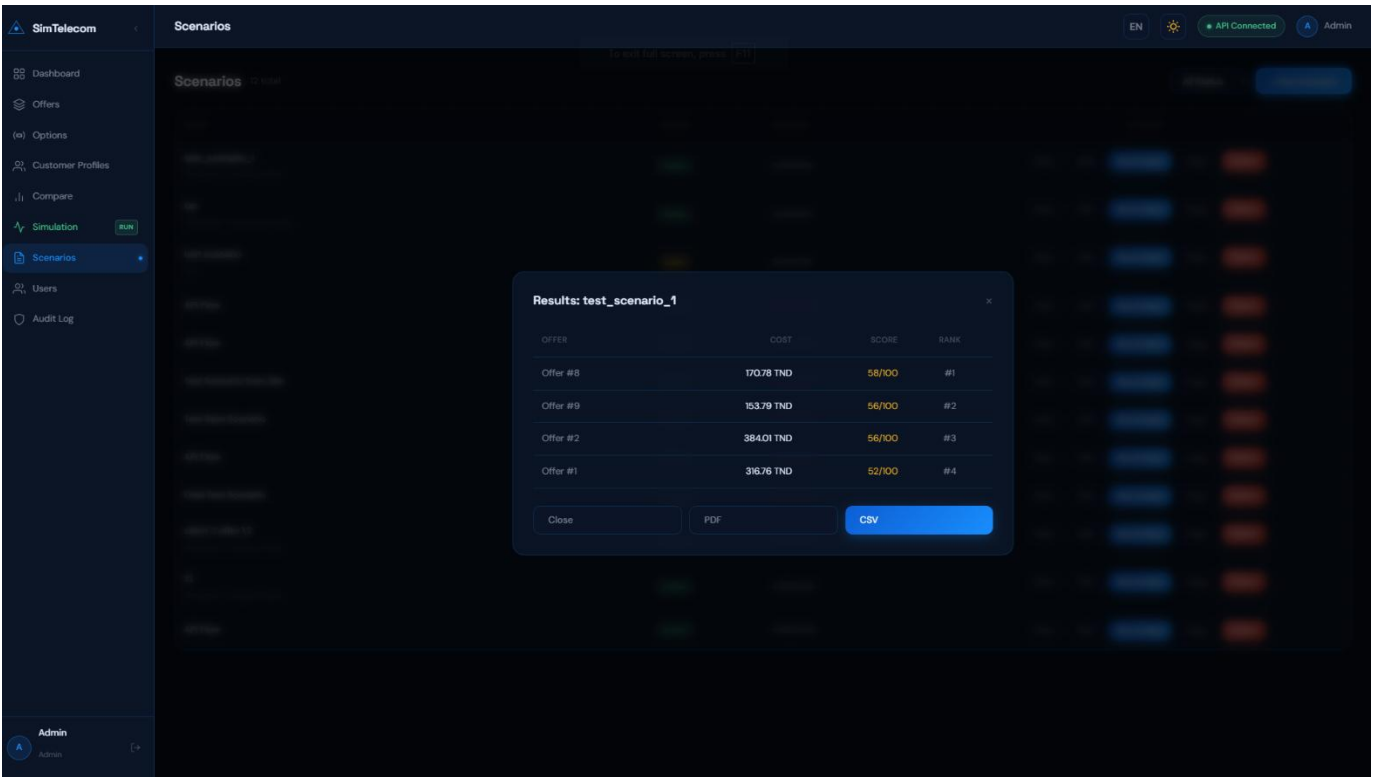


Figure 5.6 Scenario Details

This interface provides a detailed view of a selected scenario, including all associated simulation results and configurations

- Export Functionality

CHAPTER 5: Sprint 3 Scenario Management and Data Export

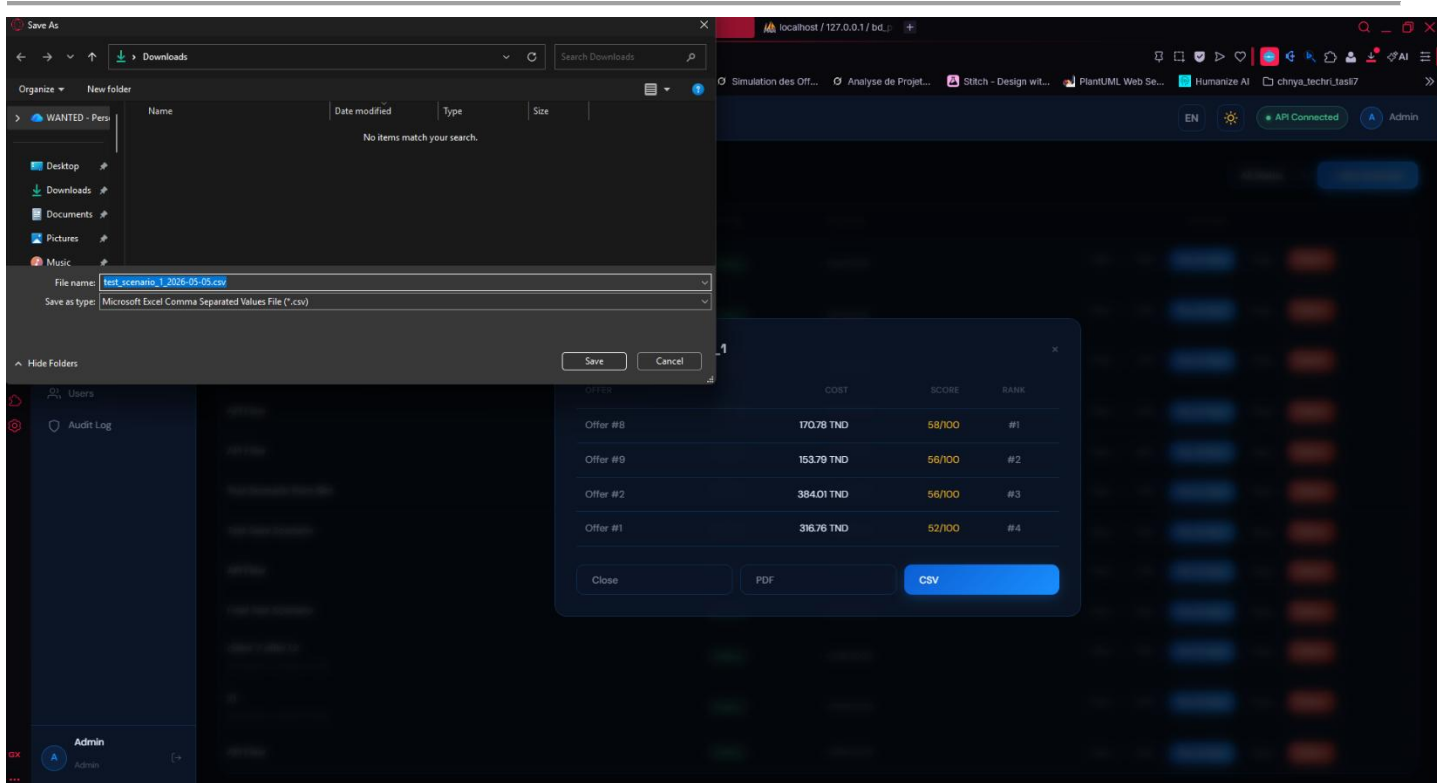


Figure 5.7 Export Results

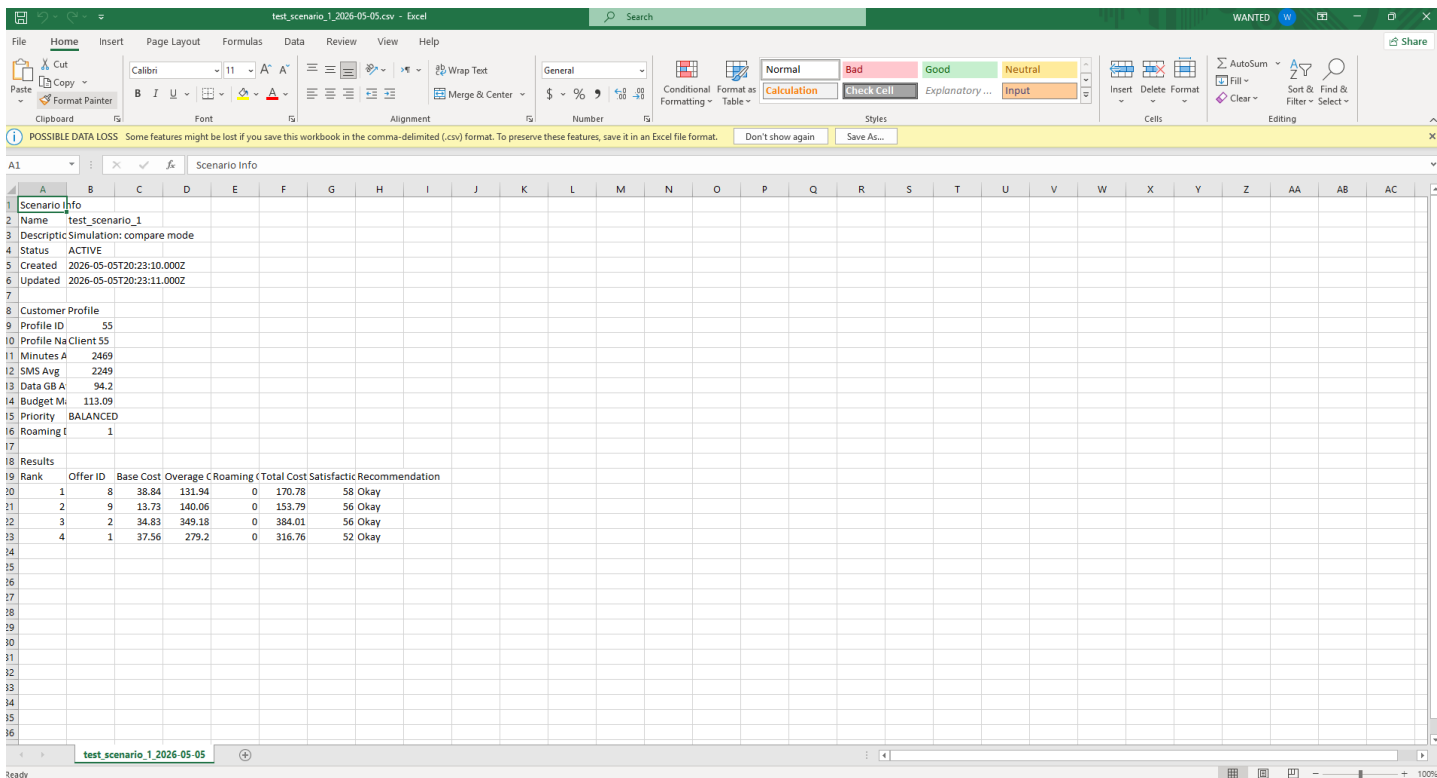


Figure 5.8 Export Results

This interface allows users to export simulation results into external file formats such as CSV or Excel. This feature facilitates further analysis and integration with Business Intelligence tools.

5.6 Testing and Validation

Testing was conducted to ensure the reliability and correctness of scenario management and export functionalities.

Table 5.5 testing and validation

Test Case	Description	Expected Result	Actual Result	Status
Save Scenario	Save simulation as scenario	Scenario stored in DB	Stored correctly	Passed
View Scenarios	Display saved scenarios list	List displayed	List displayed	Passed
Update Scenario	Modify scenario name/status	Changes saved	Changes saved	Passed
Delete Scenario	Remove a scenario	Scenario deleted	Deleted	Passed
Duplicate Scenario	Copy existing scenario	New copy created	Created successfully	Passed
Export CSV	Export results to CSV	File downloaded	File generated	Passed
Export XLSX	Export results to XLSX	File downloaded	File generated	Passed
Audit Log	Verify action is logged	Log entry created	Entry created	Passed

All functionalities were successfully tested, confirming the robustness of the scenario management module.

Conclusion

Sprint 3 enhanced the platform by introducing scenario management and export capabilities, making it more practical and aligned with real-world business needs. The addition of audit logging ensures full traceability. These improvements prepare the system for advanced BI analytics in Sprint 4.

CHAPTER 6: Sprint 4 – Business Intelligence and Data Visualization

Summary

Introduction	75
6.1 Sprint 4 Objectives.....	75
6.2 Sprint 4 Backlog.....	75
6.3 System Design	76
6.3.1 Use Case Diagram	76
6.3.2 Data Flow and Architecture.....	76
6.3.3 Detailed Use Case Descriptions.....	78
6.4 Implementation	79
6.4.1 Data Preparation.....	79
Data Sources	79
Data Processing.....	79
6.4.2 Power BI Integration.....	79
Integration Methods.....	79
Key Features.....	79
6.4.3 Dashboard Design	80
Main Dashboards.....	80
Visualization Types.....	80
6.5 Realisation.....	81
6.5.1 Application Dashboard (Internal Analytics)	81
6.5.2 Power BI Dashboard – Marketing Analysis	82
6.5.3 Power BI Dashboard – Product Analysis.....	83
6.5.4 Power BI Dashboard – Direction and Recommendation.....	83
6.5.5 Interactive Features	84
6.6 Testing and Validation.....	85
Conclusion.....	85

Introduction

Sprint 4 focuses on integrating Business Intelligence capabilities into the platform through Power BI and an internal application dashboard. This phase transforms raw simulation data into meaningful insights by providing interactive dashboards and visual analytics, supporting data-driven decision-making for marketing, product, and executive teams.

6.1 Sprint 4 Objectives

The main objectives of this sprint are:

- Prepare and structure simulation data for BI analysis (fact_simulations table, BI views)
- Build an internal application dashboard with real-time KPIs
- Create Power BI dashboards (Marketing, Product, Direction)
- Enable interactive filtering and drill-down capabilities

6.2 Sprint 4 Backlog

Table 6.1 Sprint 4 backlog

ID	User Story	Task Description	Type	Priority
12.1	Application Dashboard	Build internal dashboard with KPIs	Full-Stack	High
12.2	BI Data Preparation	Create fact_simulations table and 5 BI views	Back-End	High
12.3	BI API	Create /api/bi/* endpoints for BI data	Back-End	High
12.4	Power BI Marketing	Design marketing analysis dashboard	BI	High
12.5	Power BI Product	Design product performance dashboard	BI	High

12.6	Power BI Direction	Design executive recommendation dashboard	BI	High
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6.3 System Design

6.3.1 Use Case Diagram

This diagram illustrates user interaction with the Business Intelligence module.

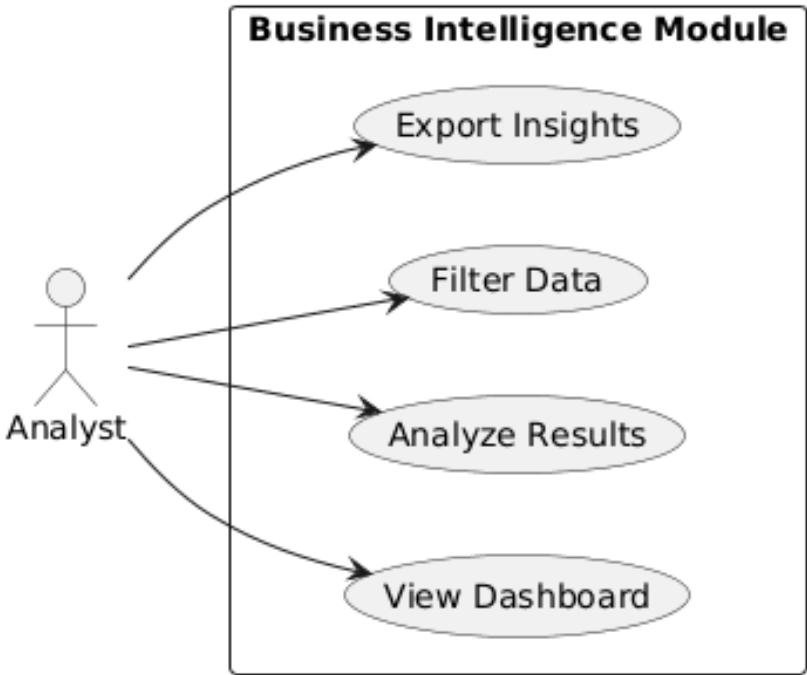


Figure 6.1 Sprint 4 Use Case Diagram

6.3.2 Data Flow and Architecture

The BI data flow follows this pipeline:

- Simulation data is generated by the engine and stored in fact_simulations table.

- Five database views aggregate this data: vw_arpu, vw_overage, vw_recommendation_rate, vw_daily_volume, vw_segment_summary.
- The /api/bi/* endpoints expose aggregated data to the internal dashboard.
- Power BI Desktop connects to exported CSV/XLSX data or directly to MySQL.
- Power BI processes data using DAX measures and renders interactive dashboards.

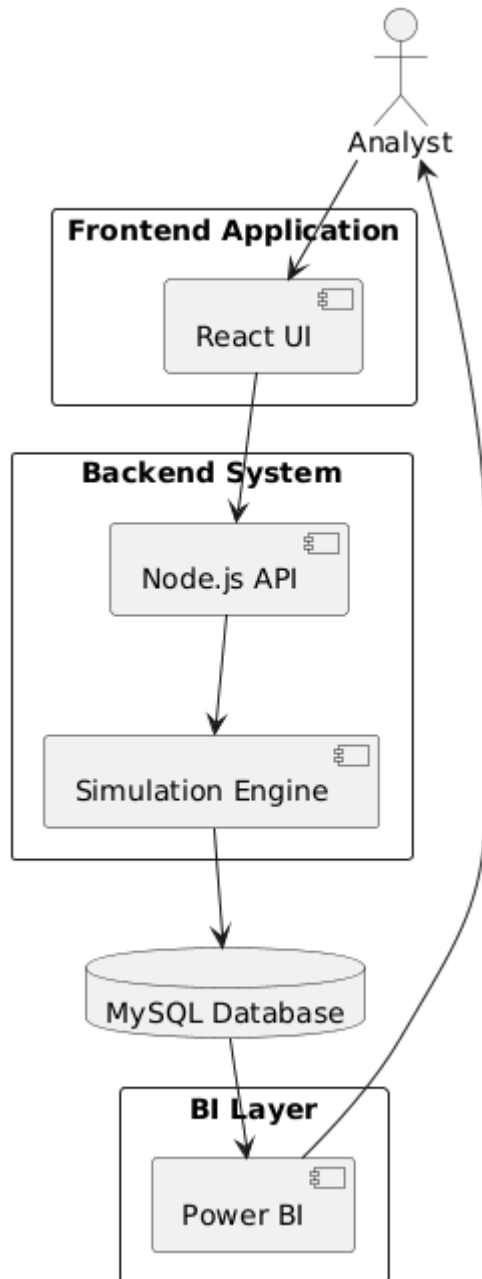


Figure 6.2 Data Flow Diagram

6.3.3 Detailed Use Case Descriptions

Use Case 1: View Application Dashboard

Table 6.2 use case 1 view dashboard

Use Case	View Application Dashboard
Actors	Analyst, Administrator
Pre-condition	The user must be authenticated. Simulation data must exist.
Nominal Scenario	1. The user navigates to the Dashboard page. 2. The system fetches real-time KPIs from /api/stats and /api/bi/*. 3. The dashboard displays: active offers count, customer profiles count, simulations run, average satisfaction score. 4. Charts show: cost trends, satisfaction trends, offers by segment, recent activity.
Post-condition	Dashboard is displayed with current data.

Use Case 2: Analyze Results in power BI

Table 6.3 use case 2 analyze Results

Use Case	Analyse Results in Power BI
Actors	Analyst, Administrator
Pre-condition	Simulation data must have been exported or Power BI must be connected to the database.

Nominal Scenario	1. The user opens Power BI Desktop. 2. Power BI loads the data model with DAX measures. 3. The user selects a dashboard (Marketing, Product, or Direction). 4. The user applies filters (segment, date range, offer type). 5. Charts and KPIs update dynamically.
Post-condition	Insights are displayed for decision-making.

6.4 Implementation

6.4.1 Data Preparation

Before integrating with Power BI, data was structured and prepared.

Data Sources

- Simulation results
- Customer profiles
- Telecom offers

Data Processing

- Cleaning and formatting data
- Structuring tables for analysis
- Ensuring consistency and integrity

6.4.2 Power BI Integration

Power BI was used to connect to the data source and build dashboards.

Integration Methods

- Direct connection to database
- Import of exported files (CSV/Excel)

Key Features

- Real-time data visualization
- Interactive filtering

- Drill-down capabilities

6.4.3 Dashboard Design

Several dashboards were created to support decision-making:

Main Dashboards

- Offer Performance Dashboard
Shows cost efficiency and usage patterns
- Customer Profile Analysis
Displays consumption trends
- Simulation Results Dashboard
Compares offers and highlights best options

Visualization Types

- Bar charts
- Pie charts
- Line graphs
- Tables

6.5 Realisation

6.5.1 Application Dashboard (Internal Analytics)

This section presents the dashboards developed both within the application and using Power BI, illustrating how simulation data is transformed into actionable insights.

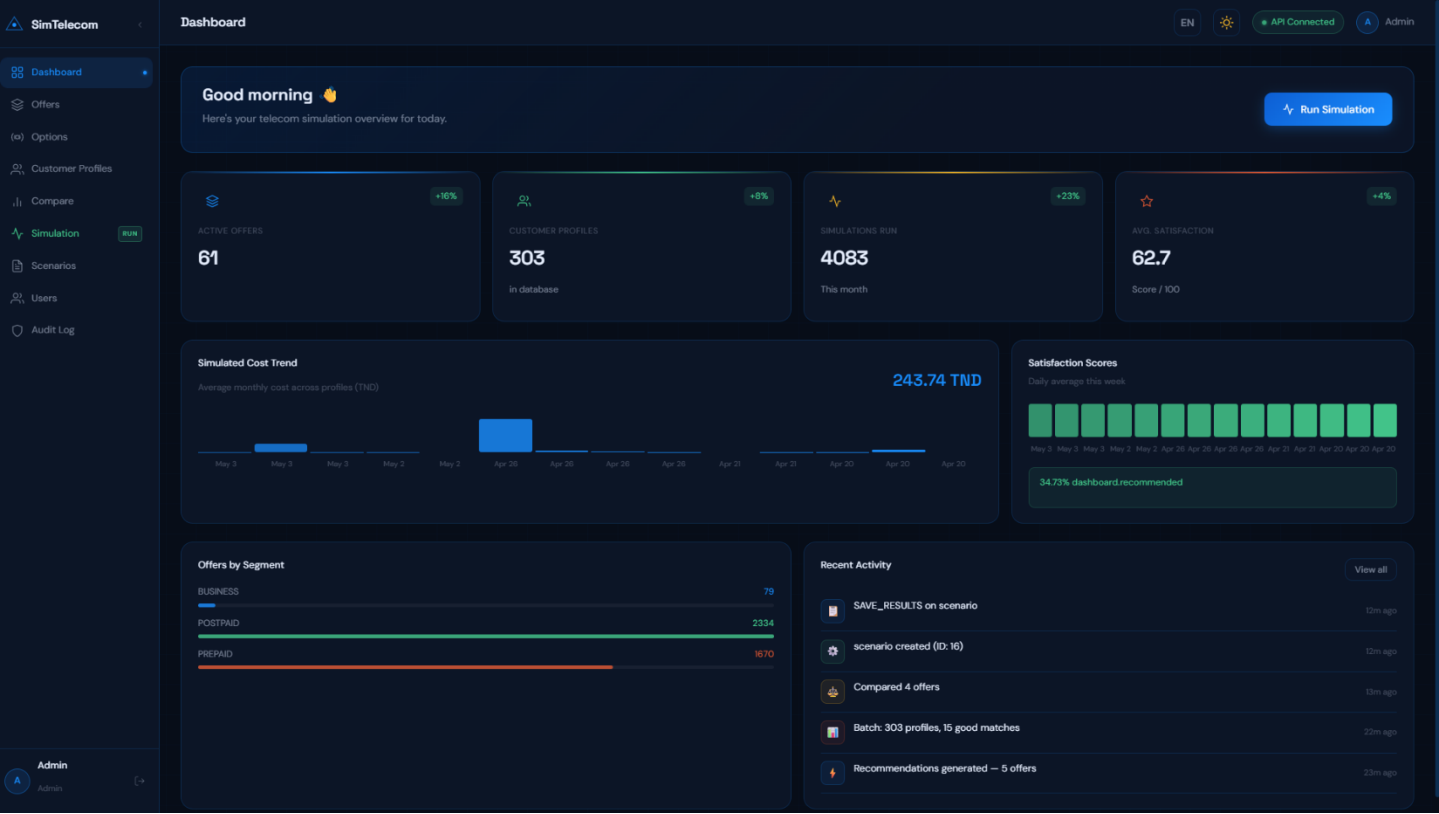


Figure 6.3 Application Dashboard

The internal dashboard provides real-time insights including active offers count, customer profiles, simulations run, average satisfaction score, simulated cost trends, satisfaction score trends, offers by segment distribution, and recent activity feed.

6.5.2 Power BI Dashboard – Marketing Analysis

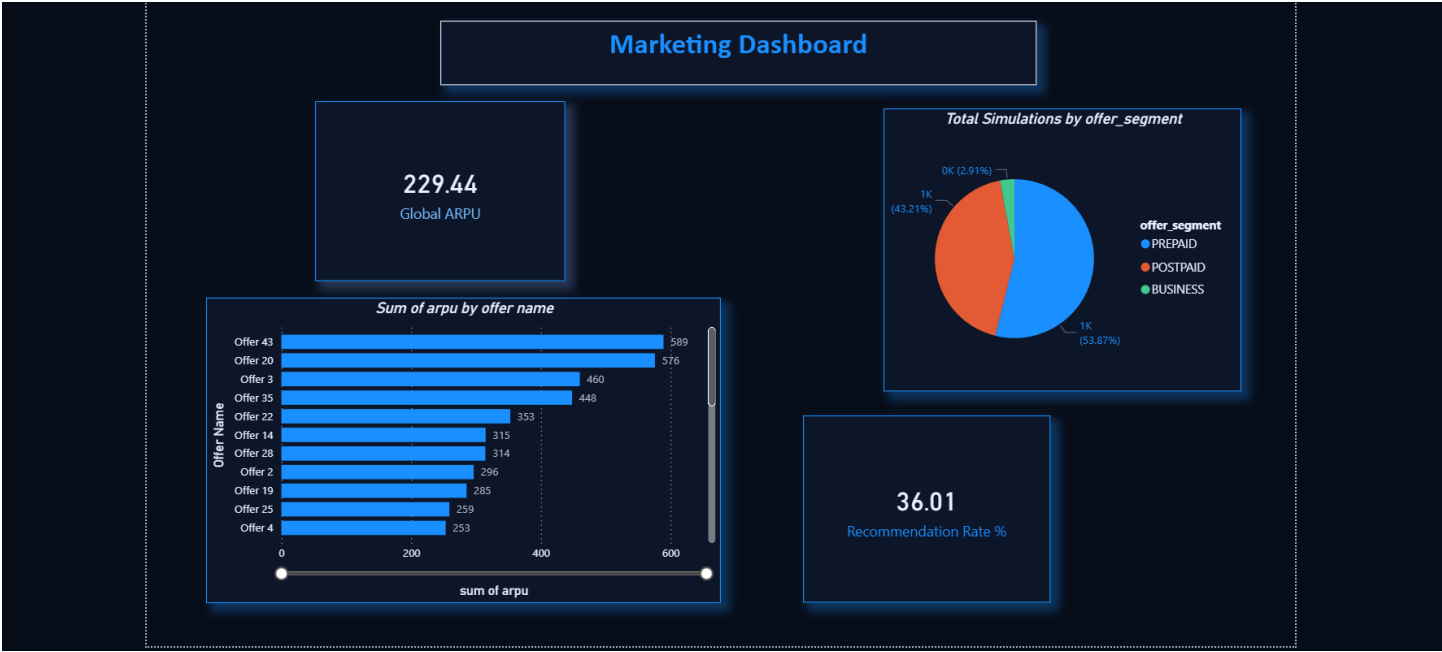


Figure 6.4 Power BI Marketing Dashboard

This dashboard focuses on marketing insights, analyzing customer segments and usage patterns. It helps identify trends in consumption and evaluate how different customer groups interact with telecom offers.

6.5.3 Power BI Dashboard – Product Analysis

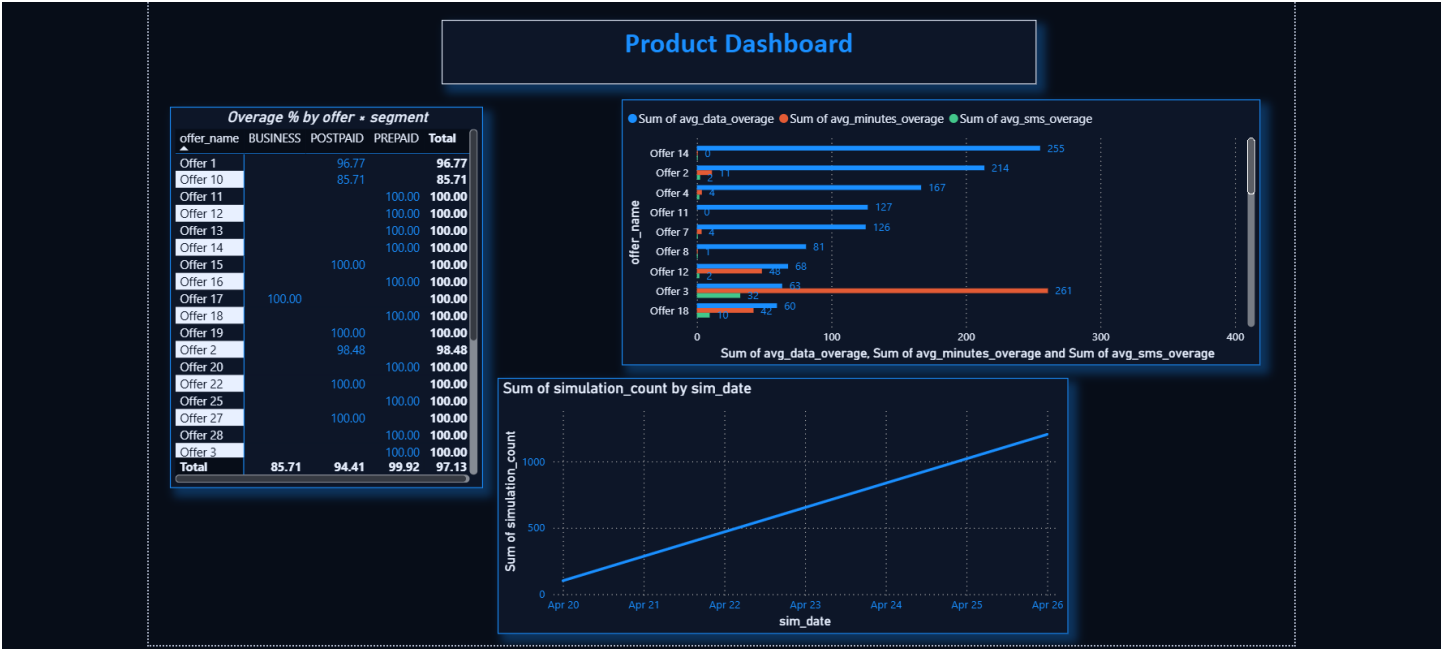


Figure 6.5 Power BI Product Dashboard

This dashboard evaluates the performance of telecom offers, highlighting cost efficiency, usage behavior, and overall effectiveness. It supports product optimization and strategic planning.

6.5.4 Power BI Dashboard – Direction and Recommendation



Figure 6.6 Power BI Direction Dashboard

This dashboard provides high-level insights and recommendations based on simulation data. It helps decision-makers identify the most suitable offers and supports strategic decision-making.

6.5.5 Interactive Features



Figure 6.7 Power BI Filtering

The dashboards include interactive features such as filtering and drill-down capabilities, allowing users to explore data dynamically and gain deeper insights.

6.6 Testing and Validation

Testing was conducted to ensure the accuracy, performance, and usability of the Business Intelligence dashboards.

Table 6.4 testing and validation

Test Case	Description	Expected Result	Actual Result	Status
Dashboard Load	Open application dashboard	KPIs load correctly	Loaded successfully	Passed
Data Accuracy	Verify dashboard data matches DB	Consistent data	Accurate data	Passed
Power BI Load	Open Power BI dashboards	All charts render	Rendered correctly	Passed
Filtering	Apply segment filter in Power BI	Data updates dynamically	Works correctly	Passed
Drill-Down	Explore detailed offer data	Detailed view displayed	Works correctly	Passed

All dashboards were successfully tested, confirming their reliability and effectiveness in supporting data analysis.

Conclusion

Sprint 4 successfully integrated Business Intelligence capabilities into the platform. The use of Power BI enables advanced data analysis and supports informed decision-making, significantly enhancing the value of the system.

GENERAL CONCLUSION AND PERSPECTIVES

General Conclusion

This graduation project focused on the design and development of a Telecom Offers Simulation and Business Intelligence Platform, carried out within the context of an end-of-studies internship at SOTETEL. The main objective was to build a decision-support system capable of analysing telecom offers based on customer usage profiles and providing meaningful insights through simulation and data visualisation.

Throughout this project, a structured and iterative approach based on the SCRUM methodology was adopted. The development process was divided into four sprints, each addressing a specific set of functionalities, ensuring progressive enhancement of the system.

The initial phase (Sprint 0) was dedicated to requirements analysis, system modelling, and architecture design. This phase laid a solid foundation by clearly defining system needs and selecting appropriate technologies.

Sprint 1 focused on implementing the core functionalities: user management, telecom offers management, options management, and customer profiles management. A robust backend and responsive frontend were established.

Sprint 2 introduced the most critical component: the simulation engine with four modes (single, compare, recommend, batch). This transformed the platform into a true decision-support tool with accurate cost calculations and satisfaction scoring.

Sprint 3 enhanced the system by adding scenario management, data export capabilities (CSV/XLSX), and comprehensive audit logging for traceability.

Sprint 4 integrated Business Intelligence functionalities using Power BI, enabling the visualisation of data through interactive dashboards for marketing, product, and executive teams.

From a technical perspective, the project successfully combined modern technologies: Node.js/Express for the backend (12 route modules), React 19/Vite for the frontend (11 pages), MySQL for data storage (9 tables, 5 BI views, 300+ customer profiles, 61 offers), and Power BI for analytics. The platform demonstrates scalability, usability, and efficiency, meeting all functional and non-functional requirements defined in the Cahier des Charges.

This project provided valuable experience in full-stack development, system design, and data analysis, while addressing a real-world problem in the telecommunications domain.

Perspectives and Future Work

- Integration of Artificial Intelligence: Implement machine learning algorithms for more advanced and personalised offer recommendations based on user behaviour and historical data.
- Real-Time Data Processing: Enable real-time data synchronisation between the system and external data sources.
- Advanced Analytics: Incorporate predictive analytics and trend analysis within the BI module.
- Mobile Application: Develop a mobile version of the platform to increase accessibility.
- Enhanced Security: Implement multi-factor authentication (MFA) and migrate tokens to httpOnly cookies.
- Integration with Telecom Systems: Connect the platform with real operator billing systems for live data.
- Production Deployment: Dockerise the application and set up a CI/CD pipeline for automated deployment.
- Database Hardening: Add foreign key constraints at the database level and implement comprehensive input validation.

These perspectives highlight the potential evolution of the platform into a more advanced, intelligent, and widely usable system capable of supporting complex decision-making processes in the telecommunications sector.